QATAR CENTRAL BANK

2025

Qatar Central Bank

QMP Interface Specifications Document V1.7.3

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1 Introduction

QMP System provides the ability for Payment Service Providers (PSPs) to communicate with the system for the purposes of sending and receiving mobile payments, and for the purpose of managing the registration of PSP customers and their accounts.

The interface specification describes at a technical level the communication of data between the PSPs and the QMP, allowing the PSPs to prepare their internal payment systems to support the interface specifications. Any specific customization is out of this document scope.

Data exchanged between the PSPs and the QMP that does not strictly match the format specified in this document is rejected. Similarly, QMP delivers the data to the PSPs in messages that are strictly formatted in accordance with the published specifications.

1.1 Scope

The requirements document covers the API specifications of QMP System available for payment service provides (PSPs). Only the specifications that have been mentioned in this document will be delivered and implemented.

1.2 Definitions and Acronyms

The below are the list of terms used throughout the document.

Term	Description
QCB	Qatar Central Bank.
Payer	The person or institution from which the payment is made.
Beneficiary	The person or institution to which the payment must be made.
PSP	Payment Service Provider. The institution responsible for providing mobile payment services for its customers.

Sender PSP	The PSP of the sending customer -the payer.
Receiver PSP	The PSP of the receiving customer - the beneficiary.
RTGS	Real-Time Gross Settlement System
Route Code	A code identifying the association between a Payment Service Provider (PSP), and a Bank. This code is used by the system to identify to which PSP is a sender of payment belongs and a receiver of payment belongs.
MAS	Mobile Account Selector; a value assigned by a PSP to link a customer account to a mobile number. Using MAS will facilitate defining multiple accounts numbers for the same mobile number.
GCC	Gulf Cooperation Council
RSP	Requirements Specification Phase.

2 Solution Overview

The below diagram provides an overview of the QMP solution, describing the role of the QMP system as switching, clearing and regulating system that facilitates exchanging mobile payment transactions between PSPs, as well as submission of NCP and total transactions to RTGS.

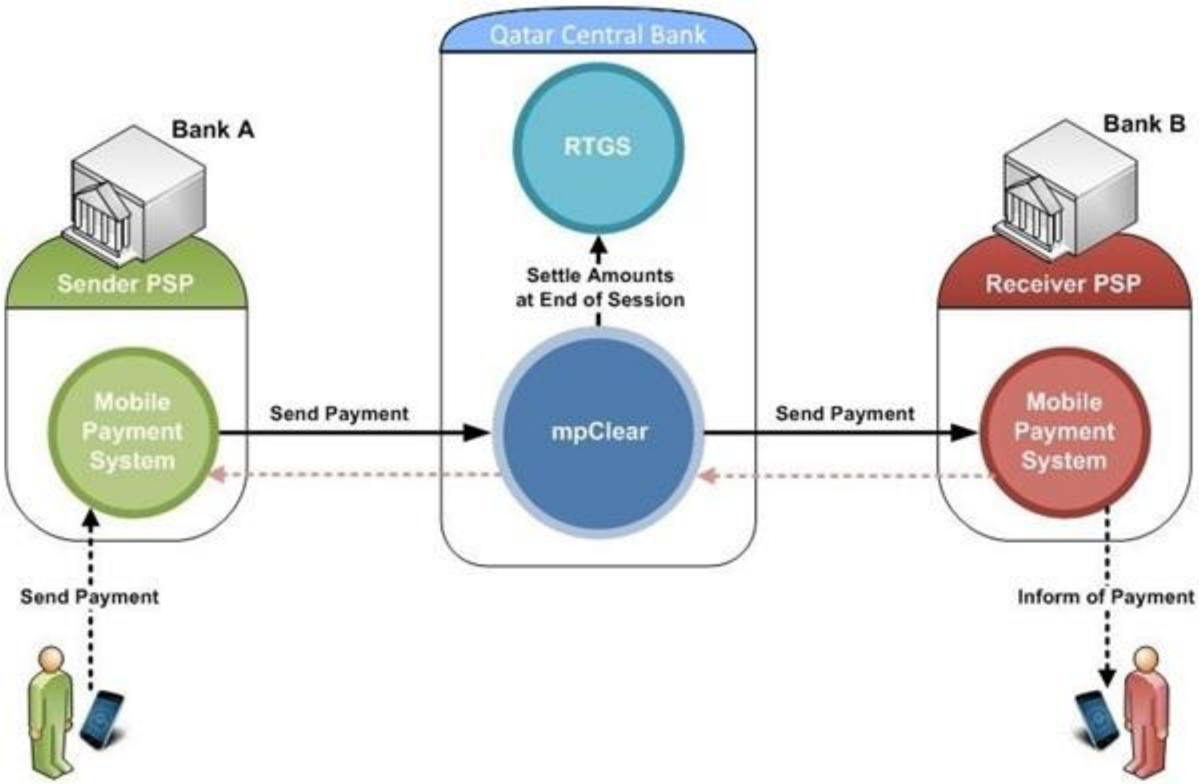


Figure: Solution Overview

As illustrated, the diagram consists of the following components:

- QMP System
- Sender PSP Mobile Payment System
- Receiver PSP Mobile Payment System

The QMP System provides switching, regulatory and clearing between PSPs. Mobile payment transactions created by PSPs are submitted through QMP System to other PSPs. The net positions of PSPs are settled at the Qatar Central Bank (QCB) RTGS through posting NCPs by QMP System.

In addition to its switching and clearing roles, QMP System also provides regulatory and operational controls, which govern the manner in which the system will operate for all PSPs. This includes -among others- the definition of clearing sessions' schedules, billing profiles, the used currencies and the allowed purposes for sending/receiving funds.

Payment Service Providers (PSPs) communicate with the QMP for the purpose of exchanging mobile payment transactions. Mobile payment transactions are initiated by a payer through his mobile phone, prompting his PSP (the sender PSP) to create a new mobile payment transaction and submit it to QMP system. In turn, QMP validates the received message and forwards it to the receiver PSP, which in turn informs its customer -the beneficiary- of receiving the funds.

In addition, PSPs communicate with the QMP to register their customers at the national directory (part of the QMP system, which contains the information of all registered customers from all PSPs). QMP supports both banked and un-banked customers; for banked customers, a PSP should communicate with the bank of those customers to validate accounts and required funds for exchanging payments (In Qatar, banks act as PSPs). To support un-banked customers, PSP will provide a wallet for each customer allowing them to send direct mobile payments to other customers, and to cash-in and cash-out money out of their wallets.

QMP system provides Message Queues (MQ)as the channel of communication for PSPs to send/receive messages; accordingly, the exchange of mobile payment transactions and registration messages between PSPs through QMP will be through that channel.

3 Mobile Payment Transactions Workflows

QMP System will provide the ability for Payment Service Providers (PSPs) to communicate with the system through its standard interfaces for exchanging mobile payment transactions. PSPs will also be able to register their customers in the QMP national directory, in addition to submitting cash-in and cash-out transactions.

QMP System shall support ISO 20022 (pacs.08.01.005) format for mobile payment transactions and cash-in/cash-out transactions. For registration transactions, QMP provides a proprietary format to support registering customers and their accounts.

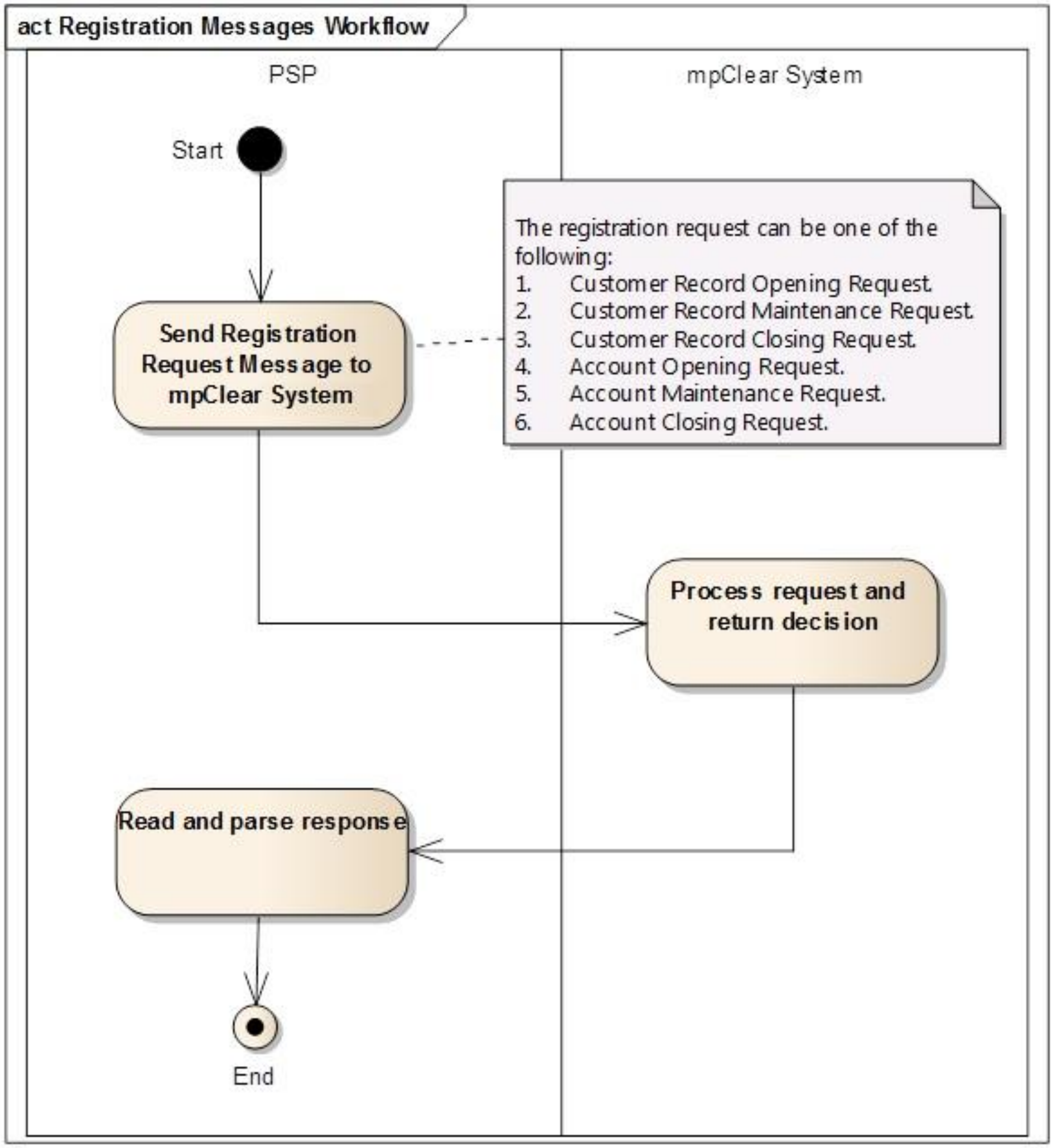
The below sections describe the workflows for mobile payment transactions.

3.1 Customer/Account Registration

One of the features provided by QMP System is the national directory, which provides a unified and centralized registry of all customers registered in the different PSPs across the country. The national directory lists the national record of each registered customer, which includes information such as the customer’s mobile number, nickname, and the bank or PSP that the customer is registered at.

Maintaining the information of registered customers is the responsibility of PSPs, which should have the facility to communicate with QMP System to manage the information of customers and their accounts in the national directory. QMP System provides a set of messages for PSPs to serve this purpose, where an instructing PSP initiates a request and sends it to QMP, while QMP processes the received message and responds with either approval or rejection.

The following workflow describes the scenario for registering a customer in QMP.



3.1.1 Registration Messages Workflow

Figure 1: Registration Messages Workflow

Workflow Overview

This workflow describes the scenario in which a PSP integrates with QMP System to manage the information of a customer/account:

- This workflow starts by the Instructing PSP preparing a registration request message. The available registration requests messages are:

1. **cstmrreg.01.01**: Customer Record Opening Request. Sent by an instructing participant to open a new record for a customer.
2. **cstmrreg.02.01**: Customer Record Maintenance Request. Sent by an instructing participant to update the information of a customer.
3. **cstmrreg.03.01**: Customer Record Closing Request. Sent by an instructing participant to close a customer record.
4. **cstmrreg.04.01**: Customer and Account Record Opening Request. Sent by an instructing participant to open a new record for a customer (and opening an associated default account).
5. **cstmrreg.06.01**: Account Opening Request. Sent by an instructing participant to open a new account for an existing customer.
6. **cstmrreg.07.01**: Account Maintenance Request. Sent by an instructing participant to update the information of a customer account.
7. **cstmrreg.08.01**: Account Closing Request. Sent by an instructing participant to close a customer account.

- The instructing PSP sends the registration message to QMP System.
- Once the message is received, QMP processes it and responds with either approval or rejection. The message sent by QMP will be **cstmrreg.10.01** (Registration Response); This message informs the instructing participant of the decision taken on a previously sent registration request

3.2 Mobile Payment Transactions

The following describes the workflow for mobile payment transactions messages initiated from Instructing PSP and submitted to the Instructed PSP through the Qatar Central Bank. An Instructing PSP will send a mobile payment message within the schedule time of a clearing session, to credit a customer of the Instructed PSP by funds debited from a customer of the Instructing PSP.

3.2.1 Mobile Payment Message Workflow

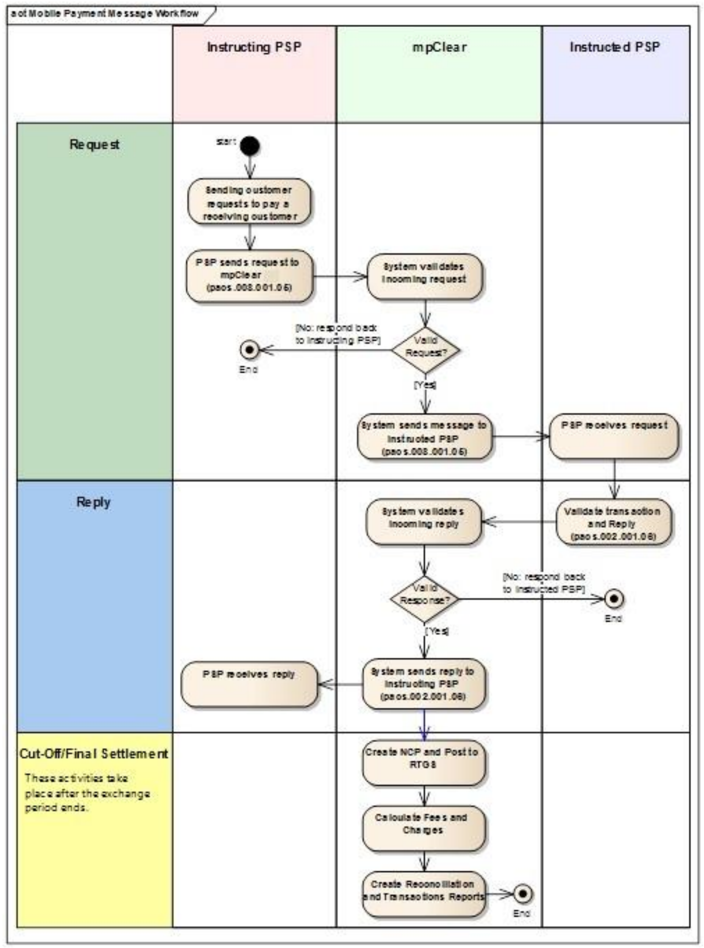


Figure 4: Mobile Payment Message Workflow

Workflow Overview

The following describes the workflow for mobile payment requests:

- A customer instructs his PSP to pay a certain amount to a beneficiary. In the request, the customer will be able to specify the mobile number (or alias) of the beneficiary along with the amount to be paid.
- Once the request is received, the Instructing PSP creates a mobile payment request.
- After creation of the request, the PSP system will send those requests in a pacs.008.001.05 message to QMP System hosted at Qatar Central Bank.
- Once QMP System receives the message, it will validate its contents and the enclosed mobile payment transaction. The validations done are described in "Message Validations" section.
- If any of the above validations above failed, QMP System will respond back to the Instructing PSP with a status report message (pacs.002.001.06), rejecting the message and detailing the reason for rejection. Otherwise (if all validations were successful), QMP will send the transaction to the related Instructed PSP.
- Once the message is received by the Instructed PSP, it will validate the mobile payment transaction and respond back to Qatar Central Bank with a status report message (pacs.002.001.06).
- Once QMP System receives the message, it will validate it. If any of the validations has failed, QMP System will respond back to the Instructed PSP that the reply is rejected, and inform the Instructing PSP that the payment has been rejected (detailing the reason for rejection). Otherwise (if all validations were successful), the system will forward the message to the Instructing PSP.
- Once the reply message is received by the Instructing PSP, it will inform its customer (the sender) with the approval (or rejection) of the transaction. On the other side, the Instructed PSP will inform its customer (the beneficiary) with the final result of processing the payment.
- If the mobile payment transaction request was approved, QMP System will change the status of the payment transaction to become "ACSP: Accepted Settlement Pending. After the exchange period of the session ends, the system will create an NCP file that contains the net positions of participants and post it to RTGS for settlement.

- After the successful settlement of NCP at RTGS, the system will calculate the fees and charges, and create the reconciliation and transactions reports accessible for the participants.

Mobile Payment Message Timeout Workflow

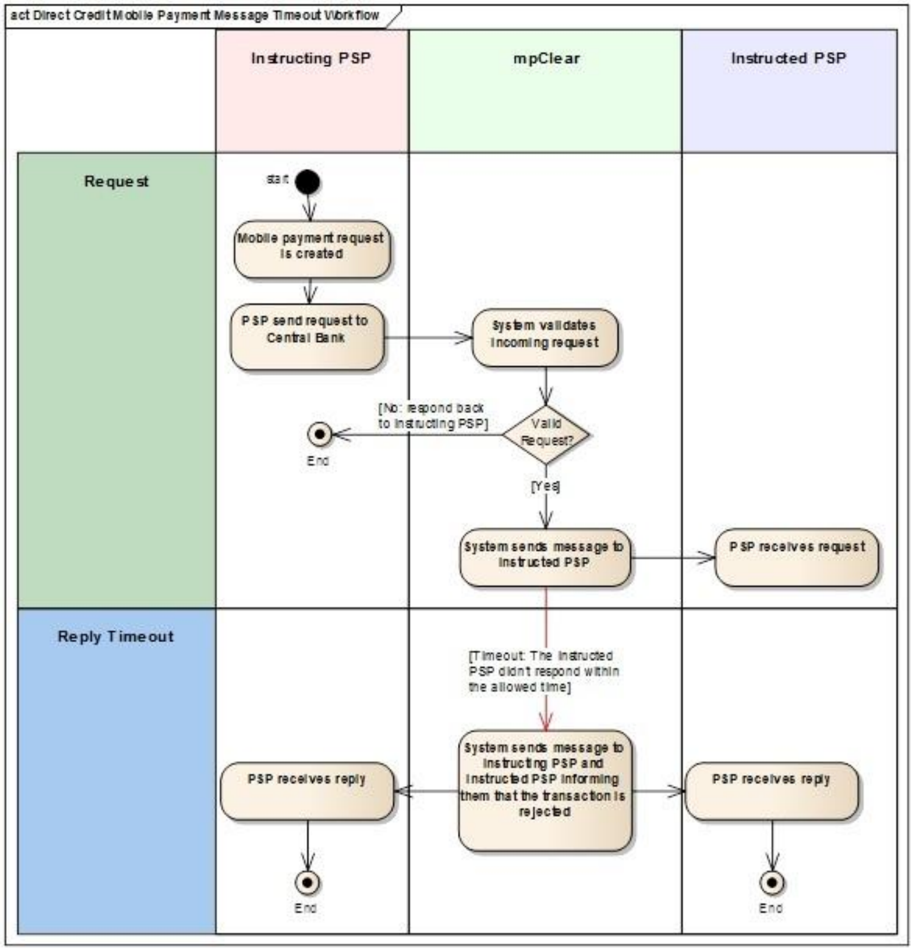


Figure 5: Direct Credit Mobile Payment Message Timeout Workflow

Workflow Overview

In case the Instructed PSP didn't respond to the received mobile payment message in the allowed time period, this will be considered as a timeout. In such a situation, QMP will send a status reply message to both the Instructing PSP and the Instructed PSP, who in turn should inform their customers with the timeout.

Notes:

- The timeout period is configurable value at QMP system. QCB users will have the ability to modify this value at any time.
- In case the Instructed PSP has responded to the mobile payment transaction after the response has timeout, the response will be rejected by QMP system, and this rejection will be sent back to the Instructed PSP.

3.2.2 Mobile Payment Message Validations

Once QMP System receives the message, it will validate its contents and the enclosed mobile payment transaction request. Among the done validations are the following:

1. Validate format of the message if it corresponds to pacs.008.001.05.
2. Validate participants defined on the transactions level, the used currency, the "mobile payment" purpose of transactions.
3. Validate that the mobile payment purpose is defined in the system. In addition, the system will validate the limits associated with the payment purpose.
4. Validate that there is an open session with presentment period that allows the provided message type according to the settlement date; the session should include an open period for presentment, with a window for the specified message type/purpose.
5. Validate the status of participants if they were active or not, and whether they are allowed to send such transactions.
6. Validate the cap limits of participants related to transactions.

In addition, the system will make the following validation pertaining to the contents of the mobile payment transaction:

1. The message consists of a single batch, and that batch has only one transaction.
2. The instructing participant and the instructed participant are defined on the batch level.
3. The debtor participant and the creditor participant are defined on the transaction level.
4. The debtor customer and the creditor customer are defined on the transaction level.
5. The debtor customer account is open (in case the customer is defined in the national directory). This validation is not supported for un-registered customers.
6. The creditor customer account is open (in case the customer is defined in the national directory). This validation is not supported for un-registered customers.
7. The provided category purpose is a valid Mobile Payment Category purpose.
8. A transaction configuration entry is defined that supports the type of debtor customer and the type of creditor customer along with the associated Mobile Payment Category Purpose.
9. The value of instructed participant field in batch should be "ACH"; QMP will know the instructed participant based to the information of creditor customer in reference to his record in the national directory.
10. The value of creditor participant field in batch should be "ACH"; QMP will know the creditor participant based to the information of creditor customer in reference to his record in the national directory.
11. The currency of batch, the currency of the creditor participant and the currency of the debtor participant should be the same.
12. The debtor customer should be registered for a route owned by the instructing participant.

3.3 Cash-In/Cash-Out

Cash-in and Cash-out transactions basically re-present movement of funds between the accounts of the customer to the cash account of the PSP; having both accounts maintained within the PSP, Cash-in and Cash-out transactions are OnUs transactions.

As regulated by Qatar Central Bank, the PSP should inform QMP System of all cash-in and cash-out transactions; Cash-in and Cash-out transactions use pacs.008.01.005 format, having the customer account in debtor part and PSP cash account in creditor part of the message (for Cash-in), or PSP cash account in debtor part and customer account in creditor part of the message (for Cash-out).

A PSP sending Cash-in or Cash-out transactions to QMP will send them as individual messages, where QMP will verify and log those transactions in its message's repository.

The following describes the workflows for cash-in and cash-out transactions.

3.3.1 Cash-In Messages Workflow

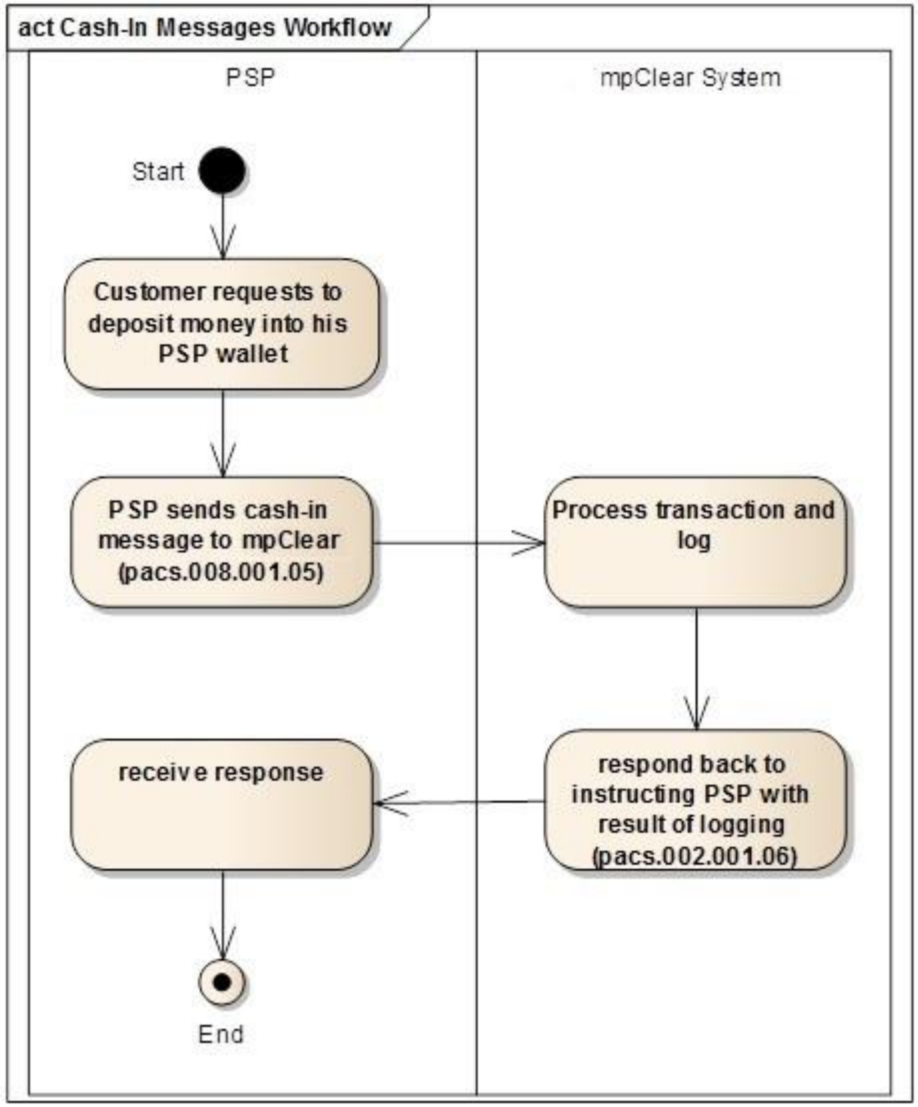


Figure 2: Cash-In Messages Workflow

Workflow Overview

This workflow describes the scenario in which a PSP integrates with QMP System to inform it with a cash-in transaction:

- This workflow starts when a customer requests to deposit an amount of money in his wallet. At that point, the PSP will send a cash-in transaction (pacs.008.001.05) to QMP System.
- Once received, QMP System will validate the inward cash-in transaction, process and log it in its repository.
- QMP System will respond back to the instructing PSP with the result of logging (pacs.002.001.06).

3.3.2 Cash-Out Messages Workflow

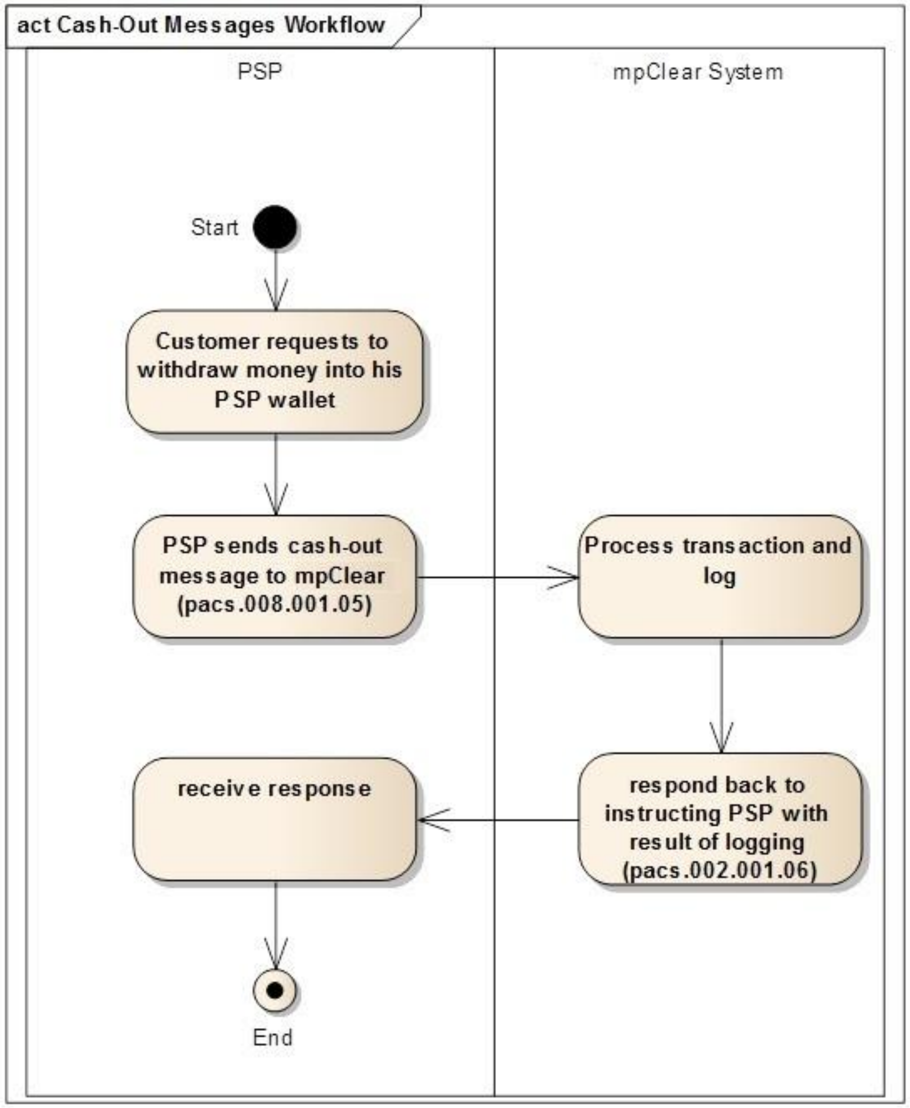


Figure 3: Cash-Out Messages Workflow

Workflow Overview

This workflow describes the scenario in which a PSP integrates with QMP System to inform it with a cash-out transaction:

- This workflow starts when a customer requests to withdraw an amount of money from his wallet. At that point, the PSP will send a cash-out transaction (pacs.008.001.05) to QMP System.
- Once received, QMP System will validate the inward cash-out transaction, process and log it in its repository.
- QMP System will respond back to the instructing PSP with the result of logging (pacs.002.001.06).

4 Registration Messages Specifications

One of the features provided by QMP System is the national directory, which provides a unified and centralized registry of all customers registered in the different PSPs across the country. The national directory lists the national record of each registered customer, which includes information such as the customer’s mobile number, nickname, and the bank or PSP that the customer is registered at.

Maintaining the information of registered customers is the responsibility of PSPs, which should have the facility to communicate with QMP System to manage the information of customers and their accounts in the national directory. QMP System provides a set of message interfaces for PSPs to serve this purpose, where an instructing PSP initiates a request and sends it to QMP, while QMP processes the received message and responds with either approval or rejection.

Payment Service providers shall be able to communicate with QMP System for the purpose of registering and maintaining the information of their customers by Registering/Maintaining the information of individual customers/accounts.

The following list of messages describes the format of registration messages exchanged between Payment Service Providers and QMP system.

4.1 *Registration Messages for Individual Customers/Accounts*

The following messages are sent by payment service providers to manage individual customers or accounts, where each message is sent to:

- Open a record for a new customer
- Maintain an existing customer record
- Close an existing customer record
- Open a record for a new account
- Maintain an existing account record
- Close an existing account record

4.1.1 Customer Record Opening Request Message

This message (identified with the code **cstmrreg.01.01**) is sent by an instructing PSP to open a new record for a customer.
This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <CstmrOpngReq>	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at the PSP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by the PSP. **The first 4 digits of the message ID should be the prefix of the bank (prefix is found in the Banks Information section).**
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Participant Identification Code	<PrtcpntId>	Identifier Set	Mandatory	Identification code of the PSP (BIC)
+ Customer Information	**<CstmrInfo>**	**Sub XML**	**Mandatory**	General information regarding the customer
++Customer Identifier	**<CstmrId>**	**Sub XML**	**Mandatory**	Customer identification information.

+++Customer Identification Number	<Id>	Text(24)	Mandatory	Identification Number of the customer (Passport number and Visa number separated by **–** in case it was a visitor)
+++Identification Type Code	<IdTp>	Text(4)	Mandatory	The code of the identification type. It should map to one of the supported identification types as defined in QMP system (found in the Identification Types section).
+++Identification Issuing Country Code	<IdIssngCtryCd>	Text(2)	Mandatory	The country that issued the ID. This field will adapt to the ISO codes for countries.
++Customer Type	<CstmrTp>	Text(10)	Mandatory	Type of the customer. It should map to one of the supported customer types as defined in QMP system (found in the Customer Types section).
++Customer Name	<CstmrNm>	Text(105)	Mandatory	Name of the customer.
++Date of Birth / Or Registration Date	<DobOrRegDt>	ISO Date	Mandatory	Date of birth for the customer (if customer is a person), or registration date of the customer (if customer is an institution).
+ Address Information	**<Address>**	**Sub XML**	**Mandatory**	Address information of the customer
++P.O. Box	<POBx>	Text(100)	Optional	Post office box of customer
++Postal Code	<PstCd>	Text(100)	Optional	Postal code of customer
++Street Name	<StrtNm>	Text(100)	Optional	Customer's street name
++Building Number	<BldgNb>	Text(100)	Optional	Customer's building number
++Phone Number	<PhneNb>	Text(100)	Optional	Customer's phone number
++Mobile Number	<MobNb>	Text(100)	Optional	Customer's mobile number
++City Name	<CtyNm>	Text(100)	Optional	Name of city
++Town Name	<TwnNm>	Text(100)	Optional	Name of town
++Governorate Name	<GvrnNm>	Text(100)	Optional	Name of governorate
++Country Code	<CtryCd>	Text(2)	Optional	The customer's country of residence. This field will adapt to the ISO codes for countries.
+ Additional Information	<AdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.

4.1.2 Customer Record Maintenance Request Message

This message (identified with the code **cstmrreg.02.01**) is sent by an instructing participant to update the information of a customer. This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <CstmrMntncReq >	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at the PSP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by the PSP. The first 4 digits of the message ID should be the prefix of the bank.
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Participant Identification Code	<PrtcpntId>	Identifier Set	Mandatory	BIC of the PSP
+ Customer Information	**<CstmrInfo>**	**Sub XML**	**Mandatory**	General information regarding the customer
++Customer Identifier	**<CstmrId>**	**Sub XML**	**Mandatory**	Customer identification information. This XML tag is needed to identify the customer whose information will be modified.
+++Customer Identification Number	<Id>	Text(24)	Mandatory	Identification Number of the customer (e.g. passport number)
+++Identification Type Code	<IdTp>	Text(4)	Mandatory	The code of the identification type. It should map to one of the supported identification types as defined in QMP system.
++Identification Issuing Country Code	<IdIssngCtryCd>	Text(2)	Mandatory	The country that issued the ID. This field will adapt to the ISO codes for countries.

++Customer Name	<CstmrNm>	Text(105)	Mandatory	Name of the customer.
++Date of Birth / Or Registration Date	<DobOrRegDt>	ISO Date	Mandatory	Date of birth for the customer (if customer is a person), or registration date of the customer (if customer is an institution).
+ Address Information	**<Address>**	**Sub XML**	**Mandatory**	Address information of the customer
++P.O. Box	<POBx>	Text(100)	Optional	Post office box of customer
++Postal Code	<PstCd>	Text(100)	Optional	Postal code of customer
++Street Name	<StrtNm>	Text(100)	Optional	Customer's street name
++Building Number	<BldgNb>	Text(100)	Optional	Customer's building number
++Phone Number	<PhneNb>	Text(100)	Optional	Customer's phone number
++Mobile Number	<MobNb>	Text(100)	Optional	Customer's mobile number
++City Name	<CtyNm>	Text(100)	Optional	Name of city
++Town Name	<TwnNm>	Text(100)	Optional	Name of town
++Governorate Name	<GvrnNm>	Text(100)	Optional	Name of governorate
++Country Code	<CtryCd>	Text(2)	Optional	The customer's country of residence. This field will adapt to the ISO codes for countries.
+ Additional Information	<AdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.

4.1.3 Customer Record Closing Request Message

This message (identified with the code **cstmrreg.03.01**) is sent by an instructing participant to close a customer record.
This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <CstmrClsgReq>	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at the PSP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by the PSP. **The first 4 digits of the message ID should be the prefix of the bank (prefix is found in the Banks Information section).**
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Participant Identification Code	<PrtcpntId>	Identifier Set	Mandatory	Identification code of the PSP
+Customer Identifier	**<CstmrId>**	**Sub XML**	**Mandatory**	Customer identification information. This XML tag is needed to identify the customer whose record will be closed.
++Customer Identification Number	<Id>	Text(24)	Mandatory	Identification Number of the customer (e.g. passport number)
++Identification Type Code	<IdTp>	Text(4)	Mandatory	The code of the identification type. It should map to one of the supported identification types as defined in QMP system.
++Identification Issuing Country Code	<IdIssngCtryCd>	Text(2)	Mandatory	The country that issued the ID. This field will adapt to the ISO codes for countries.
+ Additional Information	<AdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.

4.1.4 Customer and Account Record Opening Request Message

This message (identified with the code **cstmrreg.04.01**) is sent by an instructing PSP to open a new record for a customer, along with opening a default account associated with that customer.
This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <CstmrAndAcntOpngReq>	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at the PSP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by the PSP. **The first 4 digits of the message ID should be the prefix of the bank (prefix is found in the Banks Information section).**
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Participant Identification Code	<PrtcpntId>	Identifier Set	Mandatory	Identification code of the PSP (BIC)
+ Customer Information	**<CstmrInfo>**	**Sub XML**	**Mandatory**	General information regarding the customer
++Customer Identifier	**<CstmrId>**	**Sub XML**	**Mandatory**	Customer identification information.
+++Customer Identification Number	<Id>	Text(24)	Mandatory	Identification Number of the customer (Passport number and Visa number separated by **–** in case it was a visitor)
+++Identification Type Code	<IdTp>	Text(4)	Mandatory	The code of the identification type. It should map to one of the supported identification types as defined in QMP system (found in the Identification Types section).
+++Identification Issuing Country Code	<IdIssngCtryCd>	Text(2)	Mandatory	The country that issued the ID. This field will adapt to the ISO codes for countries.
++Customer Type	<CstmrTp>	Text(10)	Mandatory	Type of the customer.

				It should map to one of the supported customer types as defined in QMP system (found in the Customer Types section).
++Customer Name	<CstmrNm>	Text(105)	Mandatory	Name of the customer.
++Date of Birth / Or Registration Date	<DobOrRegDt>	ISO Date	Mandatory	Date of birth for the customer (if customer is a person), or registration date of the customer (if customer is an institution).
+ Address Information	**<Address>**	**Sub XML**	**Mandatory**	Address information of the customer
++P.O. Box	<POBx>	Text(100)	Optional	Post office box of customer
++Postal Code	<PstCd>	Text(100)	Optional	Postal code of customer
++Street Name	<StrtNm>	Text(100)	Optional	Customer's street name
++Building Number	<BldgNb>	Text(100)	Optional	Customer's building number
++Phone Number	<PhneNb>	Text(100)	Optional	Customer's phone number
++Mobile Number	<MobNb>	Text(100)	Optional	Customer's mobile number
++City Name	<CtyNm>	Text(100)	Optional	Name of city
++Town Name	<TwnNm>	Text(100)	Optional	Name of town
++Governorate Name	<GvrnNm>	Text(100)	Optional	Name of governorate
++Country Code	<CtryCd>	Text(2)	Optional	The customer's country of residence. This field will adapt to the ISO codes for countries.
+ Customer Additional Information	<CstmrAdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.
+Account Information	**<AcntInfo>**	**Sub XML**	**Mandatory**	Account information.
++Account Identifier	**<AcntId>**	**Sub XML**	**Mandatory**	Account identification information.
+++Mobile Number / Service Number	<MblOrSvc>	Text(35)	Mandatory	Mobile number of the account (in case the account holder is a person), Or service number of the account (in case the account holder is an institution). **Note**: the service number must be generated according to the **Merchant Identifier Structure Section**.

+++Account Type	<AcntTp>	Text(1)	Mandatory	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: service number
+++Registration Code	<RgstrtnCd>	Text(10)	Mandatory	Account registration code. A registration code is the combination of the following values: • Route Code (**Prefix** from the Banks Information Section) • MAS: Mobile Account Selector, which is a sequence number (starting from 1) assigned by the PSP to point to an actual An example of an account registration code is illustrated below, identifying with colors the above elements: **QNBA001**
++ Is Account Banked Or Unbanked	<AcntBnkd>	xs:boolean	Mandatory	Indicates whether the account is banked or unbanked. It can hold one of the following values: • true: the account is banked. • false: the account is unbanked.
++Account Currency	<Ccy>	Text(3)	Mandatory	The account's currency. This field will adapt to the ISO codes for currencies.
++Account Alias	<Alias>	Text(35)	Optional	The alias assigned to the account. The Alias regex defines a valid input as: • A string of **4 to 20 characters** in length. • Starting with a **letter** (a-z or A-Z). • Containing only **letters, digits, or underscores** (a-z, A-Z, 0-9, _). • Including at least **one digit**. • Including at least **two alphabetic characters**. Ex: Omar1@QIBA.
+ Account Additional Information	<AcntAdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.

4.1.5 Account Opening Request Message

This message (identified with the code **cstmrreg.06.01**) is sent by an instructing participant to open a new account for an existing customer. This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <AcntOpngReq>	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at the PSP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by the PSP. **The first 4 digits of the message ID should be the prefix of the bank (prefix is found in the Banks Information section).**
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Participant Identification Code	<PrtcpntId>	Identifier Set	Mandatory	Identification code of the PSP
+Customer Identifier	**<CstmrId>**	**Sub XML**	**Mandatory**	Customer identification information.
++Customer Identification Number	<Id>	Text(24)	Mandatory	Identification Number of the customer (Passport number and Visa number separated by – in case it was a visitor)
++Identification Type Code	<IdTp>	Text(4)	Mandatory	The code of the identification type. It should map to one of the supported identification types as defined in QMP system.
++Identification Issuing Country Code	<IdIssngCtryCd>	Text(2)	Mandatory	The country that issued the ID. This field will adapt to the ISO codes for countries.
+Account Information	**<AcntInfo>**	**Sub XML**	**Mandatory**	Account information.
++Account Identifier	**<AcntId>**	**Sub XML**	**Mandatory**	Account identification information.
+++Mobile Number / Service Number	<MblOrSvc>	Text(35)	Mandatory	Mobile number of the account (in case the account holder is a person), Or service number of the account (in case the account holder is an

				institution). **Note**: the service number must be generated according to the **Merchant Identifier Structure Section**.
+++Account Type	<AcntTp>	Text(1)	Mandatory	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: service number
+++Registration Code	<RgstrtnCd>	Text(10)	Mandatory	Account registration code. A registration code is the combination of the following values: • Route Code (**Prefix** from the Banks Information Section) • MAS: Mobile Account Selector, which is a sequence number (starting from 1) assigned by the PSP to point to an actual An example of an account registration code is illustrated below, identifying with colors the above elements: **QNBA001**
++ Is Account Banked Or Unbanked	<AcntBnkd>	xs:boolean	Mandatory	Indicates whether the account is banked or unbanked. It can hold one of the following values: • true: the account is banked. • false: the account is unbanked.
++Account Currency	<Ccy>	Text(3)	Mandatory	The account's currency. This field will adapt to the ISO codes for currencies.
++Account Alias	<Alias>	Text(35)	Optional	The alias assigned to the account. An alias is used to identify an account as an alternative option to mobile number. The Alias regex defines a valid input as: • A string of **4 to 20 characters** in length. • Starting with a **letter** (a-z or A-Z). • Containing only **letters, digits, or underscores** (a-z, A-Z, 0-9, _). • Including at least **one digit**. • Including at least **two alphabetic characters**. Ex: Omar1@QIBA.

+ Additional Information	<AdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.

4.1.6 Account Maintenance Request Message

This message (identified with the code **cstmrreg.07.01**) is sent by an instructing participant to update the information of a customer account. This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <AcntMntncReq>	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at the PSP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by the PSP. **The first 4 digits of the message ID should be the prefix of the bank (prefix is found in the Banks Information section).**
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Participant Identification Code	<PrtcpntId>	Identifier Set	Mandatory	Identification code of the PSP
+Account Information	**<AcntInfo>**	**Sub XML**	**Mandatory**	Account information.
++Account Identifier	**<AcntId>**	**Sub XML**	**Mandatory**	Account identification information. This XML tag is needed to identify the account whose information will be modified.
+++Mobile Number / Service Number	<MblOrSvc>	Text(35)	Mandatory	Mobile number of the account (in case the account holder is a person), Or service number of the account (in case the account holder is an institution).
+++Account Type	<AcntTp>	Text(1)	Mandatory	A character denoting the type of account. It can be one of the following values:

				• M: mobile number • C: service number
+++Registration Code	<RgstrtnCd>	Text(10)	Mandatory	Account registration code. A registration code is the combination of the following values: • Route Code • MAS: Mobile Account Selector, which is a sequence number (starting from 1) assigned by the PSP to point to an actual An example of an account registration code is illustrated below, identifying with colors the above elements:**QIBA001**
++Account Alias	<Alias>	Text(35)	Optional	The alias assigned to the account. An alias is used to identify an account as an alternative option to mobile number.
++Is Default Account	<DfltAcnt>	xs:boolean	Optional	Indicates whether the account is the default account for the customer or not. It can hold one of the following values: • true: the account is the default account. • false or "not defined": the account is not the default account.
+ Additional Information	<AdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.

4.1.7 Account Closing Request Message

This message (identified with the code **cstmrreg.08.01**) is sent by an instructing participant to close a customer account.
This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <AcntClsgReq>	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at the PSP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by the PSP. **The first 4 digits of the message ID should be the prefix of the bank (prefix is found in the Banks Information section).**
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Participant Identification Code	<PrtcpntId>	Text (35)	Mandatory	Identification code of the PSP
+Account Identifier	**<AcntId>**	**Sub XML**	**Mandatory**	Account identification information. This XML tag is needed to identify the account whose record will be closed.
++Mobile Number / Service Number	<MblOrSvc>	Text(35)	Mandatory	Mobile number of the account (in case the account holder is a person), Or service number of the account (in case the account holder is an institution).
++Account Type	<AcntTp>	Text(1)	Mandatory	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: service number
++Registration Code	<RgstrtnCd>	Text(10)	Mandatory	Account registration code. A registration code is the combination of the following values:

				• Route Code • MAS: Mobile Account Selector, which is a sequence number (starting from 1) assigned by the PSP to point to an actual An example of an account registration code is illustrated below, identifying with colors the above elements:**QIBA001**
+ Additional Information	<AdtnlInf>	Text(100)	Optional	Additional information related to the customer registration request.

4.1.8 Registration Response Message

This message (identified with the code **cstmrreg.10.01**) is sent by QMP to inform the instructing participant of the decision taken on a previously received registration request. It will hold the reply on the following requests:

- cstmrreg.01.01
- cstmrreg.02.01
- cstmrreg.03.01
- cstmrreg.04.01
- cstmrreg.06.01
- cstmrreg.07.01
- cstmrreg.08.01

This message will be in XML format, consisting of the following elements:

Message Item	<XML Tag>	Type	Presence	Description
Message Root	<Document> <RegResp>	**Sub XML**	**Mandatory**	
+ Message Identifier	**<MsgId>**	**Sub XML**	**Mandatory**	Identification reference of the message at QMP
++ Message Identification Code	<Id>	Text (35)	Mandatory	Identification code of the message as assigned by QMP
++Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory	Date and time at which the message was created.
+ Original Message Identifier	**<OrgnlMsgId>**	**Sub XML**	**Optional**	Identification reference of the request message originally submitted by the PSP
++ Message Identification Code	<Id>	Text (35)	Optional	Identification code of the message as assigned by the PSP
++Message Type	<MsgTp>	Text (14)	Optional	Type of the original message
+ Original Message Status	**<OrgnlMsgSts>**	**Sub XML**	**Mandatory**	The status of the original request message in QMP system.

++Status	<Sts>	Text(4)	Mandatory	The status of the request. It can either be one of the following values: • ACPT: the request has been accepted. • RJCT: the request has been rejected.
++Reason Code	<RsnCd>	Text(10)	Mandatory	The reason code assigned by QMP on the request. This field will adapt to one of the reason codes defined in QMP system from the Rejection Reasons section.
++Narration	<Nrtn>	Text(4000)	Optional	Narrative information provided by QMP about the response.

4.1.9 Check Default Account Message

The customer will be able to check whether his/her registered account is default one or not at the bank/PSP, this will create convenience for all customers to identify their default accounts the moment that QCB allow having multi accounts per customer in different PSPs. This message will be in XML format (cstmrreg.25.01), consisting of the following elements:

4.1.9.1 Check Default Account Request Message:

Element	XML Tag	Type	Presence	Product Spec	Description
Message Root	**<IsDefAcctReq>**	**Sub Xml**	Mandatory		
Group Header	**<GrpHdr>**	**Sub XML**	Mandatory		
+Message Idenifcation Code	**<MsgId>**	**Text (10)**	Mandatory		**First 4 characters shall be the bank prefix and the rest alphanumeric**
+Creation Date Time	**<CreDtTm>**	**ISO Date Time**	Mandatory		Date and time at which the message was created.
Instructing Participant	**<PrtcpntId>**	**Identifier Set**	Mandatory		Identification code of the PSP (BIC)

Identification Code					
Mobile Number	**<MblOrSvc>**	**Text(35)**	Mandatory		**Mobile number of the account (in case the account holder is a person), Or Merchant Id of the account (in case the account holder is an institution).**
Registration Code	**<RgstrtnCd>**	**Test(7)**	Optional		Account Registration code
Account Type	**<AcntTp>**	**Text(1)**	Conditional		A character denoting the type of account. It can be one of the following values: M: mobile number C: Merchant Id If Registration code is provided this value has to be provided.

4.1.9.2 Check Default Account Response Message:

The QMP system shall reply with one of the two options (Y/N) Based on the national directory registered in the system.
This message will be in XML format (cstmrreg.26.01), consisting of the following elements:

Element	XML Tag	Type	Presence	Product Spec	Description
Message Root	**<IsDefAcctRes>**	**Sub Xml**	Mandatory		
Group Header	**<GrpHdr>**	**Sub XML**	Mandatory		
Message Idenifcation Code	**<MsgId>**	**Text (10)**	Mandatory		First 4 characters shall be the bank prefix and the rest alphanumeric
Creation Date Time	**<CreDtTm>**	**ISO Date Time**	Mandatory		Date and time at which the message was created.

Original Message ID	**<OrgnlMsgId>**	**Sub Xml**	Mandatory		Identification reference of the request message originally submitted by the PSP
+Id	**<Id>**	**Text10**	Mandatory		Identification code of the message as assigned by the PSP
+Message Type	**< MsgTp >**	**Text (14)**	Optional		Type of the original message
Original Message Status	**<OrgnlMsgSts>**	**Sub XML**	Mandatory		The status of the original request message in PS-mpClear system.
+Status	**<Sts>**	**Text(4)**	Mandatory		The status of the request. It can either be one of the following values: • ACPT: the request has been accepted. • RJCT: the request has been rejected.
+Reason Code	**<RsnCd>**	**Text(10)**	Mandatory		The reason code assigned by PS-mpClear on the request. This field will adapt to one of the reason codes defined in PS-mpClear system.
Reply	**<Rply>**	**Text(1)**	Mandatory		**Y/N**
Default Account Registration Code	**<RgstrtnCd>**	**Test(7)**	Conditional		If reply field value was “Y” this field will be populated with the default account registration code, and if it was “N” this field will be empty.

5 ISO 20022 Payment Messages Specifications

QMPsystem supports pacs.008.001.05 payment messages and pacs.002.001.06 payment status report messages as published by the ISO 20022 standards. Messages compatible with the ISO 20022XML schema are fully supported, however, a set of fields in the ISO 20022message format are core and basic to the business processing.

The specification tables mentioned in this document lists the basic and core fields in the ISO 20022. These core fields are:

- Mandatory fields according to the ISO schema.
- Optional or Conditional fields in the ISO schema but tuned in the message to suit the internal system rules.

Other fields that do not exist in the table but exist in the full ISO20022 format can be used, however, other than the schema validation; no any internal system validations are performed on these fields.

As the system develops and other features added to the system, fields in the schema that are beyond the core and basic fields will be considered at the QMP system.

Although this document can be read in isolation, to fully understand the ISO 20022 messages format specification, readers could refer to UNIFI (ISO 20022)Payments Clearing and Settlement - Maintenance2014 - 2015, Message Definition Report - Part 2.

5.1 Direct Credit

QMPsupports the Direct Credit payment requests in XML format as per the proposed ISO pacs.008.001.05FIToFICustomerCreditTransferV05message standards from swift.It is used to move funds from a debtor account to a creditor.

Direct Credit Message Specifications:

Or	Message Item	<XML Tag>	Type	Presence	Product Specifications	Description
	Message root	**<Document><FIToFICstmrCdtTrf>**	**Sub XML**	**Mandatory**		
	Group Header	**<GrpHdr>**	**Sub XML**	**Mandatory**		Set of characteristics shared by all individual transactions included in the message.
	+Message Identifier	<MsgId>	16Text	Mandatory		Point to point reference, as assigned by the instructing party, and sent to the next party in the chain to unambiguously identify the message. The ID format is: { **Prefix** }{ **Message Reference ID** } Where: - **Prefix** is 4 characters found in Banks information section. - **Message Reference ID**: is 12 alphanumeric characters that holds a reference for the transaction at the PSP Ex: **QNBA123548456215**
	+Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory		Date and time at which the message was created.
	+Number Of Transactions	<NbOfTxs>	Max15Number	Mandatory	IC1	Number of individual transactions contained in the message. This field is always fixed to 1

	+Total Interbank Settlement Amount	<TtlIntrBkSttlmAmt>	Active Currency and Amount	Mandatory	IC2, IC3	Total amount of money moved between the instructing participant and the instructed participant.
	+Interbank Settlement Date	<IntrBkSttlmDt>	ISO Date	Mandatory		Date on which the amount of money ceases to be available to the participant that owes it andwhen the amount of money becomes available to the participant to which it is due.
	+Settlement Information	**<SttlmInf>**	**Sub XML**	**Mandatory**		Specifies the details on how the settlement of the transaction(s) between the instructing Participant and the instructed participantis completed.
	++Settlement Method	<SttlmnMtd>	Code Set	Mandatory		Method used to settle the payment instructions. This field is always fixed to "CLRG"
	++ClearingSystem	**<ClrSys>**	**Sub XML**	**Mandatory**		Specification of a pre-agreed offering between clearing agents or the channel through which the payment instruction is processed.
	+++Proprietary	<Prtry>	Code Set	Mandatory		Clearing system identification in a proprietary form. This field is always fixed to "QCB"
	+PaymentTypeInformation	**<PmtTpInf>**	**Sub XML**	**Mandatory**		**Set of elements used to further specify the type of transaction.**
	++LocalInstrument	**<LclInstrm>**	**Sub XML**	**Mandatory**		**User community specific instrument. This field is also referred to as the “Channel”.**

	+++Code[1]	<Cd>	Code Set	Mandatory		Specifies the local instrument, as published in an external local instrument code list. This field shall always be fixed to "TEL"
	++Category Purpose	**<CtgyPurp>**	**Sub XML**	**Mandatory**		**Specifies the high level purpose of the instruction based on a set of pre-defined categories from the Mobile Purposes Section.**
	+++Proprietary[2]	<Prtry>	Max35Text	Mandatory		Category purpose, in a proprietary form. This field will hold mobile payment category purposes.
	+InstructingAgent	**<InstgAgt>**	**Sub XML**	**Mandatory**		**Participant that instructs the next party in the chain to carry out the (set of) instruction(s).**
	++Financial Institution Identification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under aninternationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	Identifier Set	Optional		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".

[1] Valid Local Instruments initial list will be published by QCB in a separate external list and final list will be published before the go live date of the project

[2] Valid Category Purposes initial list will be published by QCB in a separate external list and final list will be published before the go live date of the project

	+InstructedAgent	**<InstdAgt>**	**Sub XML**	**Mandatory**		**Participant that is instructed by the previous party in the chain to carry out the (set of)instruction(s).** **As the instructed Agent might not be known to the instructing (sending) agent, this field will be set to the information of the QMP system.**
	++FinancialInstitutionIdentification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under aninternationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	Identifier Set	Optional		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)". In case this field was filled, it will contain the BIC of the QMP system as assigned by QCB. **{ QMAGQAQAMPC }**
	Credit Transfer Transaction Information	**<CdtTrfTxInf>**	**Sub XML**	**Mandatory**	**IC1**	**Set of elements providing information specific to the individual credit transfer.** **Only a single credit transfer is allowed to be included in a single message.**
	+Payment Identification	**<PmtId>**	**Sub XML**	**Mandatory**		**Set of elements used to reference a payment instruction.**

	++End To End Identification	<EndToEndId>	30Text	Mandatory		Unique identification, as assigned by the initiating party, to unambiguously identify the transaction. This identification is passed on, unchanged, throughout the entire end-to-end chain.
	++Transaction Identification	<TxId>	16Text	Mandatory		Unique identification, as assigned by the first instructing participant, to unambiguously identify thetransaction that is passed on, unchanged, throughout the entire interbank chain. This field shall hold the same value defined in the field <MsgId>
	+Inter Bank Settlement Amount	<IntrBkSttlmAmt>	Active Currency and Amount	Mandatory	IC3	Amount of money moved between the instructing participant and the instructed participant.
	+Charge Bearer	<ChrgBr>	Code Set	Mandatory		Specifies which party/parties will bear the charges associated with the processing of thepayment transaction. This field will always be fixed to "SLEV"
	+Debtor	**<Dbtr>**	**Sub XML**	**Mandatory**		**Party that owes an amount of money to the (ultimate) creditor.**
	++Name	<Nm>	Max105Text	Optional		Name by which a party is known and which is usually used to identify that party.
	++Identification	**<Id>**	**Sub XML**	**Optional**		**Unique and unambiguous identification of a party.**
	+++PrivateIdentification	**<PrvtId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a person, e.g., passport.**

	++++Other	**<Othr>**	**Sub XML**	**Optional**		**Unique identification of a person, as assigned by an institution, using an identification scheme.**
	+++++Identification	<Id>	Max24Text	Mandatory		Unique and unambiguous identification of a person/company.
	+++++Scheme Name	**<SchmeNm>**	**Sub XML**	**Optional**		**Name of the identification scheme.**
	++++++Proprietary[3]	<Prtry>	Max4Text	Mandatory		Name of the identification scheme, in a free text form.
	+++++Issuer	**<Issr>**	Code Set	Optional		Entity that assigns the identification. It will hold the country where the ID was issued.
	+Debtor Account	**<DbtrAcct>**	**Sub XML**	**Mandatory**		Unambiguous identification of the account of the debtor to which a debit entry will be made as a result of the transaction.
	++Identification	**<Id>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification for the account between the account owner and the account servicer.**
	+++Other	**<Othr>**	**Sub XML**	**Mandatory**		**Unique identification of an account, as assigned by the account servicer, using an identification scheme.**

[3] Valid Private Identification Schemes initial list will be published by QCB in a separate external list and final list will be published before the go live date of the project

	++++Identification	<Id>	Max34Text	Mandatory		Identification assigned by an institution. This field will hold the full mobile account of the sender. It will consist of the following: - Sender Registration Code. Which consist of 4 characters of the sender PSP route code, and Mobile Account Selector - Sender Account Type (M, C, or A) - Sender Mobile Number (in case type was M), or service number (in case type was C), or alias (incase type was A) e.g. QNBA001M0097498533444
	+Debtor Agent	**<DbtrAgt>**	**Sub XML**	**Mandatory**		**Financial institution servicing an account for the debtor.**
	++FinancialInstitutionIdentification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	Identifier Set	Mandatory		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
	+Creditor Agent	**<CdtrAgt>**	**Sub XML**	**Mandatory**		Financial institution servicing an account for the creditor. **As the creditor Agent might not be known to the instructing (sending) agent, this field will be set to the information of the QMP system.**

	++FinancialInstitutionIdentification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	Identifier Set	Optional		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)". In case this field was filled, it will contain the BIC of the QMP system as assigned by QCB.
	+Creditor	**<Cdtr>**	**Sub XML**	**Mandatory**		**Party to which an amount of money is due.** **This tag is only filled in case of OnUs payments; for inter-PSP payments, this tag will be empty**
	++Name	<Nm>	Max105Text	Optional		Name by which a party is known and which is usually used to identify that party.
	++Identification	**<Id>**	**Sub XML**	**Optional**		**Unique and unambiguous identification of a party.**
	+++PrivateIdentification	**<PrvtId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a person, e.g., passport.**
	++++Other	**<Othr>**	**Sub XML**	**Optional**		**Unique identification of a person, as assigned by an institution, using an identification scheme.**
	+++++Identification	<Id>	Max24Text	Mandatory		Unique and unambiguous identification of a person/company.
	+++++Scheme Name	**<SchmeNm>**	**Sub XML**	**Optional**		**Name of the identification scheme.**

	++++++Proprietary[4]	<Prtry>	Max4Text	Mandatory		Name of the identification scheme, in a free text form.
	+++++Issuer	**<Issr>**	Code Set	Optional		Entity that assigns the identification. It will hold the country where the ID was issued.
	+Creditor Account	**<CdtrAcct>**	**Sub XML**	**Mandatory**		Unambiguous identification of the account of the creditor to which a credit entry will be posted as a result of the payment transaction.
	++Identification	**<Id>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification for the account between the account owner and the account servicer.**
	+++Other	**<Othr>**	**Sub XML**	**Mandatory**		**Unique identification of an account, as assigned by the account servicer, using an identification scheme.**

[4]Valid Private Identification Schemes initial list will be published by QCB in a separate external list and final list will be published before the go live date of the project

	++++Identification	<Id>	Max34Text	Mandatory		Identification assigned by an institution. This field will hold the full mobile account of the receiver. It will consist of the following: - Receiver Registration Code. Which consist of 4 characters of the receiver PSP route code, and Mobile Account Selector - Receiver Account Type (M, C,or A) - Receiver Mobile Number (in case type was M), or service number (in case type was C), or alias (in case type was A) e.g. QNBA001M0097498533444 Note: if the registration code of the receiver was not known for the sending PSP, it will be replaced with "XXXXXXX"
	Supplementary Data	**<SplmtryData>**	**Sub XML**	**Mandatory**		**Additional information that cannot be captured in the structured elements and/or any other specific block.**
	+Place and Name	<PlcAndNm>	Text	Mandatory		This field shall be fixed to: "ACHSupplementaryData"
	+Envelope	**<Envlp>**	**SubXML**	**Mandatory**		
	++ACH Supplementary Data	**<**ns2:achSupplementaryData>	**SubXML**	**Mandatory**		**Contains supplementary data for the transaction**

	+++Batch Source	<ns2:batchSource>	Number(2)	Mandatory		Contains the batch source of the transaction. This field shall always be filled with the value "2".
	+++Consumer Id	<ns2:consumerID>	Text(16)	Optional		Consumer Id is an additional data a merchant can send and that is related to the customer transactions. This field should be populated from Consumer Id field from QR Data.
	+++MerchantCategoryCode	<ns2:merchCatego ryCd>	Number(4)	Optional		This field is mandatory for all merchant Presented QR Code transaction and should be populated from Merchant Category Code field from QR Data.
	+++ GroupMerchId	<ns2:grpMerchId>	Text(16)	Optional		This optional data element can be used to populate Group merchant Id for merchants that multiple business under common legal name This field should be populated from the GroupMerchId data in the QR code.
	+++TerminalId	<ns2:terminalId>	Text (16)	Optional		Terminal Identification is a unique code identifying the terminal at the merchant location. This field should be populated from the TerminalId data in the QR code.
	+++Filler	<ns2:filler>	Text(30)	Optional		Reserved for future use.
	+++MerchantName	<ns2:merchantNa me>	Text (20)	Optional		establishment name in case of merchant Presented QR Code transaction This field should contain Beneficiary name in case of Person to Person payments. This field should contain Business

	+++Point of Initiation Method	<ns2:pntOfInitiate Methd>	Number(2)	Optional		This field should be populated from the Point of Initiation Method data in the QR code.
	+++Tip or Convenience Indicator	<ns2:tipOrConvnce Indctr>	Number(2)	Optional		This field should be populated with data form field "Tip or Convenience Indicator" in QR data
	+++ Value of Tip or Convenience fee amount	<ns2:feePercentag e>	Decimal	Optional	E.g. 9.000 or 9.230	The Tip (from customer app) or Convenience fee amount if any available should be populated in this field. Please note transaction amount in "TtlIntrBkSttlmAmt" should be sum of transaction amount and "Value of Tip or Convenience fee amount" if available.
	+++ Country Code	<ns2:countryCd>	Text(2)	Optional		This field should be populated with data from "Country Code" field in QR data and should be in line with ISO 3166.
	+++MerchantCity	<ns2:merchantCity >	Text (15)	Optional		This field is mandatory for all merchant Presented QR Code transaction, This field should be populated from the Merchant City data in the QR code.
	+++Postal Code	<ns2:postCd>	Text(10)	Optional		This field is mandatory for all merchant Presented QR Code transaction This field should be populated from the Postal Code data in the QR code.
	+++InvoiceNumber	<ns2:invoiceNumb er>	Number(20)	Optional		The invoice No issued by the merchant This field should be populated from the Invoice No data in the QR code.

	+++SessionSequence	<ns2:sessionSequence>	Number(6)	Mandatory		When banks submit the credit message then the value should be defaulted to “0”. In case of receiving Inward credit message, the actual session sequence will be assigned automatically by the system

1.2.1 Constraints

IC1: NumberOfTransactionsAndCreditTransfersRule
"Group Header/Number Of Transactions" must equal the number of occurrences of "Credit Transfer Transaction Information".

IC2: TotalInterbankSettlementAmountAndSumRule
If "Group Header/Total Interbank Settlement Amount" is present, then it must equal the sum of all occurrences of "Credit Transfer Transaction Information/Interbank Settlement Amount".

IC3: TotalInterbankSettlementAmountRule
If "Group Header/Total Interbank Settlement Amount" is present, then all occurrences of "Credit Transfer Transaction Information/Interbank Settlement Amount" must have the same Payments Clearing and Settlement currency as the currency of "Group Header/Total Interbank Settlement Amount".

Note: When QMP receives the original pacs.008 message from the instructing PSP, it will create a different reference number. This reference number is used by QMP for all outgoing messages from its end to other PSPs (whether for pacs.008 or pacs.002). The instructed PSP should use the same reference number (taken from the pacs.008 messagereceived from QMP) in the response message to QMP.

5.2 Payment Return (Refund)

QMP supports the payment return requests in XML format as per the proposed ISO pacs.004.001.05PaymentReturnV05message standards from swift. It is used to undo a previously settled payment, debiting the creditor participant and crediting the debtor participant. This message is initiated from the creditor participant.

Payment Return Message Specifications:

Or	Message Item	<XML Tag>	Type	Presence	Product Specifications	Description
	Message root	<Document><FIToFICstmrDrctDbt>	Sub XML	Mandatory		
	Group Header	**<GrpHdr>**	**Sub XML**	**Mandatory**		Set of characteristics shared by all individual transactions included in the message.
	+Message Identifier	<MsgId>	16Text	Mandatory		Point to point reference, as assigned by the instructing party, and sent to the next party in the chain to unambiguously identify the message. The ID format is: { **Prefix** }{ **Message Reference ID** } Where: - **Prefix** is 4 characters found in Banks information section. - **Message Reference ID**: is 12 alphanumeric characters that holds a reference for the transaction at the PSP Ex: **QNBA123548456215**
	+Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory		Date and time at which the message was created.
	+Number Of Transactions	<NbOfTxs>	Max1Number	Mandatory		Number of individual transactions contained in the message. This value always should be 1

	+Interbank Settlement Date	<IntrBkSttlmDt>	ISO Date	Mandatory		Date on which the amount of money ceases to be available to the participant that owes it and when the amount of money becomes available to the participant to which it is due.
	+Settlement Information	**<SttlmInf>**	**Sub XML**	**Mandatory**		Specifies the details on how the settlement of the transaction(s) between the instructing participant and the instructed participant is completed.
	++Settlement Method	<SttlmnMtd>	Code Set	Mandatory		Method used to settle the (batch of) payment instructions.
	++Clearing System	<ClrSys>	Sub XML	Mandatory		Specification of a pre-agreed offering between clearing agents or the channel through which the payment instruction is processed.
	+++Proprietary	<Prtry>	TEXT(3)	Mandatory		Clearing system identification in a proprietary form. This field is always fixed to "ACH"
	+Instructing Agent	**<InstgAgt>**	**Sub XML**	**Mandatory**		**Participant that instructs the next party in the chain to carry out the (set of) instruction(s).**
	++Financial Institution Identification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	Identifier Set	Mandatory		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".

	+Instructed Agent	**<InstdAgt>**	**Sub XML**	**Mandatory**		**Participant that is instructed by the previous party in the chain to carry out the (set of) instruction(s).**
	++Financial Institution Identification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	Identifier Set	Mandatory		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
	Original Group Information	**<OrgnlGrpInfAndSts>**	**Sub XML**	**Mandatory And Single**		**Original group information concerning the group of transactions, to which the return message refers to.**
	+Original Message Identification	**<OrgnlMsgId>**	16Text	**Mandatory**		**Point to point reference, as assigned by the original instructing party, to unambiguously identify the original message.**
	+Original Message Name Identification	**<OrgnlMsgNmId>**	**Max35Text**	**Mandatory**		**Specifies the original message name identifier to which the message refers.**
	Transaction Information	**<TxInf>**	**Sub XML**	**Mandatory and Single**		**Provides information on the original transactions to which the return request message refers.**
	+Return Identification	<RtrId>	16Text	Mandatory		Unique and unambiguous identifier of a return request, as assigned by the assigner. Transaction ID format will be the same as the message ID in this message.
	+Original Transaction Identification	<OrgnlTxId>	16Text	Mandatory		Unique identification, as assigned by the original first instructing participant, to unambiguously Identify the transaction.

	+Returned Inter Bank Settlement Amount	<RtrdIntrBkSttlmAmt>	Active Currency and Amount	Mandatory		Amount of money moved between the instructing participant and the instructed participant.
	+Return Reason Information	**<RtrRsnInf>**	**Sub XML**	**Mandatory**		**Provides detailed information on the return reason.**
	++Reason	**<Rsn>**	**Sub XML**	**Mandatory**		**Specifies the reason for the return from the Return Reasons section.**
	+++Proprietary[5]	<Prtry>	Max35Text	Mandatory		Reason for the return request, in a proprietary form
	++Additional Information	<AddtlInf>	Max105Text	Mandatory		Further details on the return request reason.
	SupplementaryData	**<SplmtryData>**	**SubXML**	**Mandatory**		**(Message Level)** Additional information that cannot be captured in the structured elements within the transaction.
	+Envelope	**<Envlp>**	**SubXML**	**Mandatory**		Technical element wrapping the supplementary data.
	++Supplementary Data	<ns2:supplementaryData>	SubXML	Mandatory		
	+++Items	<ns2:Items>	SubXML	Mandatory		
	++++Item	<ns2:Item>	XML Tag	Mandatory and Multiple		This tag will hold a supplementary information item. The representation of information is done in a KEY-VALUE fashion, where the KEY is represented as an attribute of the <ns2:Item> element, and the VALUE is the value of the <ns2:Item> element.
	batchSource	<ns2:Item key="batchSource">	Number(2)	Mandatory		Contains the code of the source of the batch (message) as defined in the ACH system.

[5] List of Return Reasons initial list will be published by QCB in a separate external list and final list will be published before the go live date of the project

5.3 Payment Status Report

QMP supports the Payment Status Report in XML format as per the proposed pacs.002.001.06FIToFIPaymentStatusReportV06 message standards from swift.It is used to inform this party about the positive or negative status of an instruction (either single or file).

Payment Status Report Message Specifications:

Or	Message Item	<XML Tag>	Type	Presence	Product Specifications	Description
	Message root	**<Document><FIToFIPmtStsRpt>**	**Sub XML**	**Mandatory**		
	Group Header	**<GrpHdr>**	**Sub XML**	**Mandatory**		**Set of characteristics shared by all individual transactions included in the message.**
	+Message Identifier	<MsgId>	16Text	Mandatory		Point to point reference, as assigned by the instructing party, and sent to the next party in the chain to unambiguously identify the message. The ID format is: { **Prefix** }{ **Message Reference ID** } Where: - **Prefix** is 4 characters found in Banks information section. - **Message Reference ID**: is 12 alphanumeric characters that holds a reference for the transaction at the PSP Ex: **QNBA123548456215**
	+Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory		Date and time at which the message was created.
	+InstructingAgent	**<InstgAgt>**	**Sub XML**	**Mandatory**		**Participant that instructs the next party in the chain to carry out the (set of) instruction(s).**

	++Financial Institution Identification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under aninternationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	[Identifier Set]()	Optional		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
	+InstructedAgent	**<InstdAgt>**	**Sub XML**	**Mandatory**		**Participant that is instructed by the previous party in the chain to carry out the (set of) instruction(s).** **This field will consist of the instructing agent of the original message.**
	++FinancialInstitutionIdentification	**<FinInstnId>**	**Sub XML**	**Mandatory**		**Unique and unambiguous identification of a financial institution, as assigned under aninternationally recognized or proprietary identification scheme.**
	+++BICFI	<BICFI>	[Identifier Set]()	Optional		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
	Original Group Information And Status	**<OrgnlGrpInfAndSts>**	**Sub XML**	**Mandatory And Single**	**[PC1]()**	**Original group information concerning the group of transactions, to which the status report message refers to.**
	+OriginalMessageIdentification	<OrgnlMsgId>	16Text	Mandatory		Point to point reference, as assigned by the original instructing party, to unambiguously identify the original message.

	+OriginalMessageNameIdentification	<OrgnlMsgNmId>	Max35Text	Mandatory		Specifies the original message name identifier to which the message refers. (e.g. pacs.008.001.05)
	+Group Status	<GrpSts>	Code Set	Mandatory	IC5	Specifies the status of a group of transactions.
	+Status Reason Information	**<StsRsnInf>**	**Sub XML**	**Conditional**	IC5	**Provides detailed information on the status reason.If it was Accepted then no need to set a reason code**
	++Reason	**<Rsn>**	**Sub XML**	**Mandatory**		**Specifies the reason for the status report from the Rejection reasons section.**
	+++Proprietary[6]	<Prtry>	Max35Text	Mandatory		Reason for the status, in a proprietary form.
	++Additional Information	<AddtlInf>	Max105Text	Optional		Further details on the status reason.
	Supplementary Data	**<SplmtryData>**	**Sub XML**	**Mandatory**		**Additional information that cannot be captured in the structured elements and/or any other specific block.**
	+Place and Name	<PlcAndNm>	Text	Mandatory		This field shall be fixed to: "ACHSupplementaryData"
	+Envelope	**<Envlp>**	**SubXML**	**Mandatory**		
	++ACH Supplementary Data	**<ns2:achSupplementaryData>**	**SubXML**	**Mandatory**		**Contains supplementary data for the transaction**
	+++Batch Source	<ns2:batchSource>	Number(2)	Mandatory		Contains the batch source of the transaction. This field shall always be filled with the value "20".
	+++receiverIdIssuingCountry	<ns2:receiverIdIssuingCountry>	Code Set	Optional		The country of the issuer.
	+++receiverIdType	<ns2: receiverIdType>	Max4Text	Optional		Type code of the ReceiverId

[6] Valid Reasons initial list will be published by QCB in a separate external list and final list will be published before the go live date of the project

	+++receiverIdValue	<ns2:receiverIdValue>	Max24Text	Optional		Value of the Receiver Id
	+++receiverName	<ns2:receiverName>	Max105Text	Optional		Name of the Receiver.
	+++SessionSequence	<ns2:sessionSequence>	Number(6)	Mandatory		This field should always contain the clearing session sequence and should match the original message session sequence.

5.3.1 Constraints

IC5: Status Reason Information Rule
If "Group Status" is present and is different from RJCT then "Status Reason Information/Additional Information" must be absent.

PC1: Original Group Information And Status Single Presence Rule
"Original Group Information And Status" must be present and only once.

5.4 Debit Request Technical Specifications

QMP supports the Direct Debit payment requests in XML format as per the proposed ISO pacs.003.001.05FIToFICustomerDebitTransferV05 message standards from swift.It is used to collect funds from a debtor account for a creditor.

This type of messages shall enable scenarios where a beneficiary (e.g. merchant) is requesting to be credited with a certain amount from a customer, requiring the beneficiary to initiate a transaction from their systems (e.g. POS), keying in the mobile number of the customer along with the requested amount.
Accordingly, merchants should be able to initiate the payment transaction, and request their customer to authorize the payment using customer bank provided Mobile app

Direct Debit Message Specifications:

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
Group Header	**<GrpHdr>**	**Sub XML**	**Mandatory**		Set of characteristics shared by all individual transactions included in the message.
+Message Identifier	<MsgId>	16Text	Mandatory		Point to point reference, as assigned by the instructing party, and sent to the next party in the chain to unambiguously identify the message. The ID format is: { **Prefix** }{ **Message Reference ID** } Where: - **Prefix** is 4 characters found in Banks information section. - **Message Reference ID**: is 12 alphanumeric characters that holds a reference for the transaction at the PSP Ex: **QNBA123548456215**
+Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory		Date and time at which the message was created.
+Number Of Transactions	<NbOfTxs>	Max15Number	Mandatory	IC1	Number of individual transactions contained in the message.
+Total Interbank Settlement Amount	<TtlIntrBkSttlmAmt>	Active Currency and Amount	Mandatory	IC2, IC3	Total amount of money moved between the instructing participant and the instructed participant. This should be populated with data

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					form field Transaction currency and transaction amount of the QR data. Please note the amount in TtlIntrBkSttlmAmt should be the sum of the actual transaction amount and any other surcharges or TIP if applicable.
+Interbank Settlement Date	<IntrBkSttlmDt>	ISO Date	Mandatory		Date on which the amount of money ceases to be available to the participant that owes it and when the amount of money becomes available to the participant to which it is due.
+Settlement Information	**<SttlmInf>**	**Sub XML**	**Mandatory**		Specifies the details on how the settlement of the transaction(s) between the instructing Participant and the instructed participant is completed.
++Settlement Method	<SttlmnMtd>	Code Set	Mandatory	PC1	Method used to settle the (batch of) payment instructions. This field is always fixed to "CLRG"
++ClearingSystem	<ClrSys>	Sub XML	Mandatory		Specification of a pre-agreed offering

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					between clearing agents or the channel through which the payment instruction is processed.
+++Proprietary	<Prtry>	Code Set	Mandatory		Clearing system identification in a proprietary form. This field is always fixed to "QMP"
+PaymentTypeInformation	**<PmtTpInf>**	**Sub XML**	**Mandatory**		Set of elements used to further specify the type of transaction.
++LocalInstrument	**<LclInstrm>**	**Sub XML**	**Optional**		User community specific instrument.
+++Code[7]	<Cd>	Code Set	Mandatory		Specifies the local instrument. It will always be defined as TEL
++CategoryPurpose	**<CtgyPurp>**	**Sub XML**	**Mandatory**		Specifies the high level purpose of the instruction based on a set of pre-defined categories.
+++Proprietary[8]	<Prtry>	Number(3)	Mandatory		Category purpose, in a proprietary form.
+InstructingAgent	**<InstgAgt>**	**Sub XML**	**Mandatory**		Participant that instructs the next party in the chain to carry out the (set of) instruction(s).

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
++Financial Institution Identification	**<FinInstnId>**	**Sub XML**	**Mandatory**		Unique and unambiguous identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.
+++BICFI	<BICFI>	Identifier Set	Mandatory		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
+InstructedAgent	**<InstdAgt>**	**Sub XML**	**Conditional**		Participant that is instructed by the previous party in the chain to carry out the (set of) instruction(s). As the instructed Agent might not be known to the instructing (sending) agent, this field will be set to the information of the QMP system.
++FinancialInstitutionIdentification	**<FinInstnId>**	**Sub XML**	**Mandatory**		Unique and unambiguous identification of a financial institution, as assigned under an internationally

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					recognized or proprietary identification scheme.
+++BICFI	<BICFI>	Identifier Set	Mandatory		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
Direct Debit Transaction Information	**<DrctDbtTxInf>**	**Sub XML**	**Mandatory And Multiple**	IC1	Set of elements providing information specific to the individual direct debit(s).
+Payment Identification	**<PmtId>**	**Sub XML**	**Mandatory**		Set of elements used to reference a payment instruction.
++End To End Identification	<EndToEndId>	Max35Text	Mandatory		Unique identification, as assigned by the initiating party, to unambiguously identify the transaction. This identification is passed on, unchanged, throughout the entire end-to-end chain. This field should be populated with Merchant Txn Identifier.Merchant Ref”

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					from the QR Data or a transaction reference number.
++Transaction Identification	<TxId>	16Text	Mandatory		Unique identification, as assigned by the first instructing participant, to unambiguously identify the transaction that is passed on, unchanged, throughout the entire interbank chain. This field shall hold the same value defined in the field <MsgId>
+Inter Bank Settlement Amount	<IntrBkSttlmAmt>	Active Currency and Amount	Mandatory	IC3	Total amount of money moved between the instructing participant and the instructed participant. This should be populated with data form field Transaction currency and transaction amount of the QR data. Please note the amount in TtlIntrBkSttlmAmt should be the sum of the actual transaction amount and any other surcharges or TIP if applicable

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
+Charge Bearer	<ChrgBr>	Code Set	Mandatory	PC2	Specifies which party/parties will bear the charges associated with the processing of the payment transaction.
+Creditor	**<Cdtr>**	**Sub XML**	**Mandatory**		Party to which an amount of money is due.
++Name	<Nm>	Max105Text	Optional		Name by which a party is known and which is usually used to identify that party. This field should be populated with data from "Merchant Name/ Beneficiary Name" or field of QR Data.
++Identification	**<Id>**	**Sub XML**	**Optional**		Unique and unambiguous identification of a party.
+++PrivateIdentification	**<PrvtId>**	**Sub XML**	**Mandatory**		
++++Other	**<Othr>**	**Sub XML**	**Optional**		
+++++Identification	<Id>	Max35Text	Mandatory		Unique and unambiguous identification of a party.
+++++Scheme Name	**<SchmeNm>**	**Sub XML**	**Optional**		Name of the identification scheme
++++++Proprietary[9]	<Prtry>	Max35Text	Mandatory		Name of the identification scheme, in a free text form.

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
+++++Issuer	**<Issr>**	Code Set	Optional		Entity that assigns the identification. It will hold the country where the ID was issued.
+Creditor Account	**<CdtrAcct>**	**Sub XML**	**Mandatory**		
++Identification	**<Id>**	**Sub XML**	**Mandatory**		
+++Other	**<Othr>**	**Sub XML**	**Mandatory**		
++++Identification	<Id>	Max34Text	Mandatory		Identification assigned by an institution. This field will hold the full mobile account of the creditor. It will consist of the following: - Creditor Registration Code. Which consist of 4 characters of the creditor PSP route code, and Mobile Account Selector - Creditor Account Type (M, C,or A) - Creditor Mobile Number (in case type was M), or service number (in case type was C), or alias (in case type was A) e.g. QNBA001M0097498533444
+Creditor Agent	**<CdtrAgt>**	**Sub XML**	**Mandatory**		Financial institution servicing an account for the creditor.
++FinancialInstitutionIdentification	**<FinInstnId>**	**Sub XML**	**Mandatory**		Unique and unambiguous

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.
+++BICFI	<BICFI>	Identifier Set	Mandatory		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
+Debtor	**<Dbtr>**	**Sub XML**	**Mandatory**		Party that owes an amount of money to the (ultimate) creditor. As the debtor Agent might not be known to the instructing (sending) agent, this field will be set to the information of the QMP system.
++Name	<Nm>	Max105Text	Optional		Name by which a party is known and which is usually used to identify that party.
++Identification	**<Id>**	**Sub XML**	**Optional**		Unique and unambiguous identification of a party.
+++PrivateIdentification	**<PrvtId>**	**Sub XML**	**Mandatory**		Unique and unambiguous identification of a person, eg, passport.

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
++++Other	**<Othr>**	**Sub XML**	**Optional**		Unique identification of a person, as assigned by an institution, using an identification scheme.
+++++Identification	<Id>	Max35Text	Mandatory		
+++++Scheme Name	**<SchmeNm>**	**Sub XML**	**Optional**		Name of the identification scheme.
++++++Proprietary[10]	<Prtry>	Max35Text	Mandatory		Name of the identification scheme, in a free text form.
+++++Issuer	**<Issr>**	Code Set	Optional		Entity that assigns the identification. It will hold the country where the ID was issued.
+Debtor Account	**<DbtrAcct>**	**Sub XML**	**Mandatory**		Unambiguous identification of the account of the debtor to which a debit entry will be made as a result of the transaction.
++Identification	**<Id>**	**Sub XML**	**Mandatory**		Unique and unambiguous identification for the account between the account owner and the account servicer.
+++Other	**<Othr>**	**Sub XML**	Mandatory		Unique identification of an account, as assigned by the account servicer,

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					using an identification scheme.
++++Identification	<Id>	Max34Text	Mandatory		Identification assigned by an institution. This field will hold the full mobile account of the debtor. It will consist of the following: - Debtor Registration Code. Which consist of 4 characters of the debtor PSP route code, and Mobile Account Selector - Debtor Account Type (M, C,or A) - Debtor Mobile Number (in case type was M), or service number (in case type was C), or alias (in case type was A) e.g. QNBA001M0097498533444 Note: if the registration code of the Debtor was not known for the sending PSP, it will be replaced with "XXXXXXX"
+Debtor Agent	**<DbtrAgt>**	**Sub XML**	**Mandatory**		Financial institution servicing an account for the debtor.
++FinancialInstitutionIdentification	**<FinInstnId>**	**Sub XML**	**Mandatory**		Unique and unambiguous

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.
+++BICFI	<BICFI>	Identifier Set	Mandatory		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".
Supplementary Data	**<SplmtryData>**	**Sub XML**	**Mandatory**		**Additional information that cannot be captured in the structured elements and/or any other specific block.**
+Place and Name	<PlcAndNm>	Text	Mandatory		This field shall be fixed to: "ACHSupplementaryData"
+Envelope	**<Envlp>**	**SubXML**	**Mandatory**		
++ACH Supplementary Data	<ns2:achSupplementaryData>	**SubXML**	**Mandatory**		**Contains supplementary data for the transaction**
+++Batch Source	<ns2:batchSource>	Number(2)	Mandatory		Contains the batch source of the transaction. This field shall always be filled with the value "2".

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
+++Consumer Id	<ns2:consumerID>	Text(16)	Optional		Consumer Id is an additional data a merchant can send and that is related to the customer transactions. This field should be populated from Consumer Id field from QR Data.
+++MerchantCategoryCode	<ns2:merchCategoryCd>	Number(4)	Optional		This field is mandatory for all merchant Presented QR Code transaction and should be populated from Merchant Category Code field from QR Data.
+++ GroupMerchId	<ns2:grpMerchId>	Text(16)	Optional		This optional data element can be used to populate Group merchant Id for merchants that multiple business under common legal name This field should be populated from the GroupMerchId data in the QR code.
+++TerminalId	<ns2:terminalId>	Text (16)	Optional		Terminal Identification is a unique code identifying the terminal at the merchant location.

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					This field should be populated from the TerminalId data in the QR code.
+++Filler	<ns2:filler>	Text(30)	Optional		Reserved for future use.
+++MerchantName	<ns2:merchantName>	Text (20)	Optional		establishment name in case of merchant Presented QR Code transaction This field should contain Beneficiary name in case of Person to Person payments. This field should contain Business
+++Point of Initiation Method	<ns2:pntOfInitiateMethd>	Number(2)	Optional		This field should be populated from the Point of Initiation Method data in the QR code.
+++Tip or Convenience Indicator	<ns2:tipOrConvnceIndctr>	Number(2)	Optional		This field should be populated with data form field “Tip or Convenience Indicator” in QR data
+++ Value of Tip or Convenience fee amount	<ns2:feePercentage>	Decimal	Optional	E.g. 9.000 or 9.230	The Tip (from customer app) or Convenience fee amount if any available should be populated in this field. Please note transaction amount in “TtlIntrBkSttlmAmt” should be sum of transaction amount and “Value of Tip or

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					Convenience fee amount” if available.
+++ Country Code	<ns2:countryCd>	Text(2)	Optional		This field should be populated with data from “Country Code” field in QR data and should be in line with ISO 3166.
+++MerchantCity	<ns2:merchantCity>	Text (15)	Optional		This field is mandatory for all merchant Presented QR Code transaction, This field should be populated from the Merchant City data in the QR code.
+++Postal Code	<ns2:postCd>	Text(10)	Optional		This field is mandatory for all merchant Presented QR Code transaction This field should be populated from the Postal Code data in the QR code.
+++InvoiceNumber	<ns2:invoiceNumber>	Number(20)	Optional		The invoice No issued by the merchant This field should be populated from the Invoice No data in the QR code.
+++SessionSequence	<ns2:sessionSequence>	Number(6)	Mandatory		When banks submit the credit message then the value should be defaulted to “0”.

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
					In case of receiving Inward credit message, the actual session sequence will be assigned automatically by the system

5.4.1 Constraints

IC1: NumberOfTransactionsAndDirect DebitRule
"Group Header/Number Of Transactions" must equal the number of occurrences of "Direct Debit Transaction Information".

IC2: TotalInterbankSettlementAmountAndSumRule
If "Group Header/Total Interbank Settlement Amount" is present, then it must equal the sum of all occurrences of "Direct Debit Transaction Information/Interbank Settlement Amount".

IC3: TotalInterbankSettlementAmountRule
If "Group Header/Total Interbank Settlement Amount" is present, then all occurrences of "Direct Debit Transaction Information/Interbank Settlement Amount" must have the same Payments Clearing and Settlement currency as the currency of "Group Header/Total Interbank Settlement Amount".

IC4: Settlement Priority Rule
All occurrences of "Settlement Priority" should have the same value (being Normal or High) within the same batch.

PC1: Settlement Method Rule
"Settlement Method" is always equal to "CLRG".

PC2: ChargeBearerRule
"Charge Bearer"always contains "SLEV".

5.5 Payment Enquiry

The Payment Enquiry Request message will be in XML as per the ISO Pacs.028.001.01 FIToFIPmtStsReq, it aims to get the status of a processed payment. Such message can be initiated by either debtor or creditor bank for P2P and P2B transactions. The response to this message will be provided by QMP system and will contain the detailed status of the transaction. Payment Enquiry can be initiated by providing the data block of "+OriginalGroupInformation" highlighted in Blue or data block of "+Transaction info" highlighted in Orange in the below table.

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
Message Root	<FIToFIPmtStsReq>				
+Group Header	<GrpHdr>	Sub XML	Mandatory		
++MessageIdentification	<MsgId>	16Text	Mandatory		It should be unique and structured as follows: {Participant Prefix} {Message Reference ID} Where: - Participant Prefix is 4 characters prefix - Message Reference ID: is 12 alphanumeric characters that holds a reference for the request message.
++CreationDateTime	<CreDtTm>	ISO Date Time	Mandatory		(YYYY-MM-DDThh:mm:ss.sss+/-hh:mm)
++InstructingAgent	<InstgAgt>	Identifier Set	Optional		Participant that instructs the next party in the chain to carry out the (set of) instruction(s).
+++FinancialInstitutionIdentification	<FinInstnId>	**Sub XML**	Optional		Unique and unambiguous identification of a financial institution, as assigned under an internationally recognized or proprietary identification scheme.
++++BICFI	<BICFI>	Identifier Set	Optional		Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".

Message Item	<XML Tag>	Type	Presence	Product Spec	Description
+OriginalGroupInformation	<OrgnlGrpInf>	**Sub XML**	Conditional		Original group information concerning the group of transactions, to which the payment enquiry message refers to.
++OriginalMessageIdentification	<OrgnlMsgId>	16Text	Mandatory		The original message ID
+++OriginalMessageNameIdentification	<OrgnlMsgNmId>	Max35Text	Mandatory		The original message type (e.g. pacs.008.001.05)
++OriginalCreationDateAndTime	<OrgnlCreDtTm>	ISO Date Time	Optional		(YYYY-MM-DDThh:mm:ss.sss+/-hh:mm)
+Transaction info	<TxInf>		Conditional		Either Original Group Information or Transaction Info has to be provided.
++OriginalEndToEndIdentification	<OrgnlEndToEndId>	Max35Text	Mandatory		If the transaction info group is provided this field is mandatory
+Supplementary Data	<SplmtryData>	Sub XML	Mandatory		Additional information that cannot be captured in the structured elements and/or any other specific block.
++Place and Name	<PlcAndNm>	Text	Mandatory		This field shall be fixed to: "ACHSupplementaryData"
++Envelope	**<Envlp>**	SubXml	Mandatory		
+++ACH Supplementary Data	<ns2:achSupplementaryData>	**SubXML**	Mandatory		Contains supplementary data for the transaction
++++ **GroupMerchId**	<ns2: grpMerchId>	Text16	Mandatory		The merchant identifier is the Merchant Id to be used for initiating merchant payment and should comply with the standard defined in QMP operating rules.
++++Terminal Id	<ns2:terminalId>	Text16	Mandatory		Terminal Identification is a unique code identifying the terminal at the merchant location.

5.6 Customer Name Verification

This is to cross check the beneficiary/receiver name before processing the payment it aims to reduce the mistakes that might happen while submitting the payments and secure the debtor by ensuring that the beneficiary name match the information has been given by the receiver. The name shall be returned in clear format.

5.6.1 Customer Name Verification Request

This request message will be in XML format (cstmrreg.20.01), consisting of the following elements:

Message Item	XML Tag	Type	Presence	Product Spec	Description
Message Root	<CstmrNameReq>	Sub Xml	Mandatory		
Group Header	<GrpHdr>	Sub XML	Mandatory		
+Message Identification Code	<MsgId>	Text (10)	Mandatory		Please rerfer tosection 8.7 for details.
+Creation Date Time	<CreDtTm>	ISO Date Time	Mandatory		Date and time at which the message was created.
Instructing Participant Identification Code	<PrtcpntId>	Identifier Set	Mandatory		Identification code of the PSP (BIC)
Instructing Mobile	< InstrMblOrSvc >	Text35	Conditional		If this value was filled then Instructing Alias is optional

Instructing Alias	< InstrAlias>	Text35	Conditional		If this value was filled then Instructing mobile number is optional
Mobile Number	<MblOrSvc>	Text35	Conditional		Mobile number of the account (in case the account holder is a person), Or Merchant Id of the account (in case the account holder is an institution).If this value was filled then the Alias is optional
Alias	<Alias>	Text35	Conditional		If this value was filled then the Mobile Number is optional

Customer Type	**<CstmrTp>**	**Text(10)**	Mandatory		Customer type, valid values: 'PER', 'Person' 'CORP', 'Corporate' 'GOV', 'Government Agency',

5.6.2 Customer Name Verification Response

This message shall display the customer name based on the request message, the name shall be returned in clear format it will be in XML format (cstmrreg.21.01) consisting of the following elements:

Element	XML Tag	Type	Presence	Product Spec	Description
Message Root	**<CstmrNameRes>**	**Sub Xml**	Mandatory		
Group Header	**<GrpHdr>**	**Sub XML**	Mandatory		
+Message Identification Code	**<MsgId>**	**Text10**	Mandatory		
+Creation Date Time	**<CreDtTm>**	**ISO Date Time**	Mandatory		Date and time at which the message was created.
Original Message ID	**<OrgnlMsgId>**	**Sub Xml**	Mandatory		Identification reference of the request message originally submitted by the PSP

+Id	**<Id>**	**Text10**	Mandatory		Identification code of the message as assigned by the PSP
+Message Type	**< MsgTp >**	**Text (14)**	Optional		Type of the original message
Original Message Status	**<OrgnlMsgSts>**	**Sub XML**	Mandatory		The status of the original request message in PS-mpClear system.
+Status	**<Sts>**	**Text(4)**	Mandatory		The status of the request. It can either be one of the following values: • ACPT: the request has been accepted. • RJCT: the request has been rejected.
+Reason Code	**<RsnCd>**	**Text(10)**	Mandatory		The reason code assigned by PS-mpClear on the request. This field will adapt to one of the reason codes defined in PS-mpClear system.
Customer Name	**<CstmrNm>**	**Text(35)**	Mandatory		**Clear name shall be set in this tag**

6 Merchant Identifier Structure

The merchant identifier is also used as the service number to be used for initiating merchant payments hence, it is important that all banks adopt the same standard in defining a merchant Identifier. The merchant identifier should be constructed as follows:

First two digits will be 2-digit bank code as defined in the Institution Code + 4 digits of MCC code as defined in ISO 18245 for retail financial services + running sequence number of 10 digit.

For example:

Qatar National Bank (QNB) opening a merchant relation for Carrefour will have the below merchant identifier number:

0154111234567890

Where;

01 is the 2-digit bank code for QNB.

5411 is the MCC for the supermarkets.

1234567890 is the running sequence number.

Banks are strictly required to adhere to this standard of merchant ID creation.

7 Data Types

7.1 Active Currency And Amount

Definition:
A number of monetary units specified in an active currency where the unit of currency isexplicit and compliant with ISO 4217.

This data type contains the following XML attribute:

Name	Attribute XML Name	Data Type	Example
Currency	Ccy	Active Currency Code	<TtlIntrBkSttlmAmtCcy="QAR">100</TtlIntrBkSttlmAmt>

Format:
Total Digits: 18
Fraction Digits: 5

Note: The decimal separator is a dot.

7.2 ISODateTime

Definition:
A particular point in the progression of time defined by a mandatory date and a mandatory time component, expressed in either UTC time format withUTC offset format (YYYY-MM-DDThh:mm:ss.sss+/-hh:mm),- These representations are defined in "XML Schema Part 2: Datatypes SecondEdition - W3C Recommendation 28 October 2004" which is aligned with ISO 8601.

Example: <CreDtTm>2017-03-06T17:35:18.138+04:00</CreDtTm>

7.3 ISODate

Definition:
A particular point in the progression of time in a calendar year expressed in the YYYY-MM-DD format. This representation is defined in "XML Schema Part 2: Datatypes Second Edition - W3CRecommendation 28 October 2004" which is aligned with ISO 8601.

Example:<IntrBkSttlmDt>2015-05-26</IntrBkSttlmDt>

7.4 Local Instrument Code Set

Definition:
Specifies the external local instrument code in the format of character string with a maximumlength of 35 characters.
External code sets can be downloaded from www.iso20022.org.

7.5 BICFI Identifier Set

Definition:
Code allocated to a financial institution by the ISO 9362 Registration Authority as described in ISO 9362 "Banking - Banking telecommunication messages - Business identifier code (BIC)".

Format:
[A-Z]{6,6}[A-Z2-9][A-NP-Z0-9]([A-Z0-9]{3,3}){0,1}

Note:
Valid BICs for financial institutions are registered by the ISO 9362 Registration Authority in the BICdirectory, and consist of eight (8) or eleven (11) contiguous characters comprising the first three or all four of the following components: INSTITUTION CODE, COUNTRY CODE, LOCATION CODE, BRANCH CODE. The institution code, country code and location code are mandatory, while the branch code is optional.

7.6 Country Code Set

Definition:
Code to identify a country, a dependency, or another area of particular geopolitical interest, on the basis of country names obtained from the United Nations (ISO 3166, Alpha-2 code).

Format:
[A-Z]{2,2}

Note:
The code is checked against the list of country names obtained from the United Nations (ISO 3166, Alpha-2 code).

7.7 Payment Status Report Group Status

Definition:
Specifies the status of a group of transactions.

Code	Name	Description
RJCT	Rejected	Payment initiation or individualtransaction included in the paymentinitiation has been rejected.
ACSP	AcceptedSettlementInProcess	All preceding checks such as technicalvalidation and customer profile weresuccessful and therefore the paymentinitiation has been accepted for execution.
ACSC	AcceptedSettlementCompleted	Settlement on the debtor's account has been completed. Usage: this can be used by the first participant to report to the debtor that the transaction has been completed. Warning: this status is provided for transaction status reasons, not for financial information. It can only be used after bilateral agreement

Initiating Party Applicable Statuses Matrix:
This matrix specifies the outward statuses applicable for QMP, receiving PSPor both.

Code	QMP	Receiving PSP
RJCT	Yes	Yes
ACSP	Yes	Yes
ACSC	Yes	No

8 System HeartBeat Service

This service shall be used by the participants to check the system availability and whether their internal systems are connected to the QMP system. There is no specific message format as banks can send any text message with an empty body (no contents) and the QMP system shall send an immediate response back with system date and time following the ISO date/time format.

Note: This message is optional and it is up to the participants to use it or not.

9 Communication Method

Standard interfacing channels are supported to interface the QMP system with PSPs. The system will use Message Queue (MQ) for exchanging all registration messages and payment messages.

QCB will be hosting the Message Queue server and it will be accessible by all PSPs. The MQ Server will be Apache ActiveMQ. Each PSP will have different queues for outward and inward messages. PSPs should post their outward files (registration requests, payments requests, and payments responses) to the outbox queues. QMP will put all inward files (registration responses, payments responses, and reports) in the inbox queues.

Each participant will have the following queues hosted at the QCB message queue broker:
Note: The Short Name for each PSP is found under the Banks Information section.

- **Outward Payment Queue**: used by the participant bank to send payment messages (PACS.008,PACS.003,PACS.004,PACS.028) to QMP. Queue name shall be structured as:**mpc.{*participant_short_name*}.payment.outward**

- **Inward Payment Queue**: used by the participant bank to receive payment messages (PACS.008,PACS.003,PACS.004) from QMP.Queue name shall be structured as:**mpc.{*participant_short_name*}.payment.inward**

- **Outward Replies Queue:** used by the participant bank to send payment status report (PACS.002). Queue name shall be structured as: **mpc.{participant_short_name}.reply.outward**

- **Inward Replies Queue:** used by the participant bank to receive payment status report (PACS.002). Queue name shall be structured as**: mpc.{participant_short_name}.reply.inward**

- **Outward Registration Queue**: used by the participant bank to send individual registration messages to QMP.Queue name shall be structured as: **mpc.{*participant_short_name*}.reg.outward**
- **Inward Registration Queue**: used by the participant bank to receive individual registration responses from QMP.Queue name shall be structured as:**mpc.{*participant_short_name*}.reg.inward**
- **Outward HeartBeat Queue:** used by participant bank to send heartbeat request message. Queue name shall be structured as**: mpc.{participant_short_name}.heartbeat.outward**
- **Inward HeartBeat Queue:** used by participant bank to receive heartbeat response message. Queue name shall be structured as**: mpc.{participant_short_name}.heartbeat.inward**

When implementing the above queues for each participant bank, the phrase "***{participant_short_name}***" in the queue name is replaced with the short name of the participant bank (found in the Banks Information section) which the queues are created for. (For example, the queue with the name "**mpc.{*participant_short_name*}.payment.outward"** implemented for **Qatar National Bank** will be named **mpc.qnb.payment.outward**).

9.1 Messages Properties

The system requires receiving directive properties accompanied with the messages sent by PSPs. The received properties allow the system to handle the received message properly depending on its type. It also allows the system to authenticate the source of the message.
The type and representation of message properties depends on the type of the received message. There are two types of messages:

Non-Binary Messages: those messages associated with non-file based requests/response, which are:

- Direct Credit in MX format.
- Status Report in MX format.
- Customer Record Opening Request.
- Customer Record Maintenance Request.
- Customer Record Closing Request.

- Account Opening Request.
- Account Maintenance Request.
- Account Closing Request.
- Registration Response.

9.2 Non-Binary Messages Properties

For non-binary messages, the system expects receiving the structure of the received message to contain both the message properties and the contents of the message as XML elements. The message will consist of the following XML elements:

#	Tag	Description
1	<id>	Message Identification code
2	<type>	The type of the created message. The possible values are: • **PACS.008**: for Direct Credit in MX format. • **PACS.002**: for Status Report in MX format. • **CSTMRREG.01**: for Customer Record Opening Request. • **CSTMRREG.02**: for Customer Record Maintenance Request. • **CSTMRREG.03**: for Customer Record Closing Request. • **CSTMRREG.06**: for Account Opening Request. • **CSTMRREG.07**: for Account Maintenance Request. • **CSTMRREG.08**: for Account Closing Request. • **CSTMRREG.10**: for Registration Response.
3	<format>	The format of the message. The value is: - **MX**: for ISO20022 format **Note:** the above value is only applicable for direct credit and status report messages. For other types of messages, this field should contain empty value (i.e. should be sent as <Format></Format>).
4	<date>	Message creation date

5	<signature>	Digital signature represented in Base64 Text
6	<content>	The structured information of the message (built according to the specifications of each message as described in this document).

9.3 Messsage Correlation ID

When the participant send a message (Payment, Registration, HeartBeat), the Active MQ assigns a unique identifier to the request, when the QMP system processes the request, it saves the identifier and adds the request's identifier to the reply message (either the reply was from the receiving bank or automatically from the system).

When the sender participant processes the reply, it uses the request identifier to correlate the request with the reply. So banks are not required to send the correlation ID in their Replies messages.

10 Certificates Generation

Below are the steps that the banks need in order to request / generate the Certificates required for the digital signing of the QMP messages:

Please note that you have to change some parameters such as:

- ***CERT_ALIAS***
- ***PASSWORD***
- ***CSR_NAME***
- ***KEYSTORE_NAME***
- ***CER_NAME***

1- Generate the wallet which contains your certificate:

```
keytool -genkey -alias CERT_ALIAS -keyalg RSA -keysize 2048 -sigalg SHA256withRSA -dname "CN=QCB-MPC,OU=IT,O=QATAR CENTRAL BANK,L=DOHA,ST=DOHA,C=QA" -keypass PASSWORD -keystore KEYSTORE_NAME.jks -storepass PASSWORD
```

2- Generate a certificate request and send it to QCB:

```
keytool -certreq -v -alias CERT_ALIAS -file CSR_NAME.csr -sigalg SHA256withRSA -keypass PASSWORD -storepass PASSWORD -keystore KEYSTORE_NAME.jks
```

3- Import the Root CA received from QCB:

```
keytool -import -v -noprompt -trustcacerts -alias rootcacert -file root.cer -keystore KEYSTORE_NAME.jks -storepass PASSWORD
```

4- Import User CA received from QCB:

```
keytool -import -v -alias CERT_ALIAS -file CER_NAME.cer -keystore KEYSTORE_NAME.jks -keypass PASSWORD -storepass PASSWORD
```

5- The PSP will export the Public certificate and provide it to QCB (QMP system) in order to verify the inward messages:

```
keytool" -export -alias CERT_ALIAS -keystore KEYSTORE_NAME.jks -rfc -file CER_NAME.cer
```

After Applying the above steps, the PSP will have their own private certificate that will be used to sign the messages sent to QCB (QMP system). The PSP needs to request the QMP's public certificate in order to verify the inward messages received from the QMP system.

10.1 Digital Signature

A digital signature is a value that is generated by the sender of the message to authenticate itself to the receiver. All messages exchanged between the PSP and the QMP System should be digitally signed. PSPs will use their digital certificates issued by QCB to do the digital signing, placing the digital signature as part of the message created by the PSP system, and using the following algorithms while building the digital signature:

- **SHA-256**: the hashing algorithm used to create the hash value that will constitute the message digest.
- **RSA**: to calculate the digital signature using the above created message digest.

The signing and message preparation procedure will be as following:

1. The PSP system will construct the message token using the following fields for non-binary messages:
   a. **Message Content**: the message contents that has been generated by system of the PSP.
   b. **Secret statement**: the Secret statement is a string that is only known to the PSP and configured in QMP (Note: the current value of it is empty).
   c. **Message creation Date**: date is formatted in XML format (Note: the date should be until the seconds only, ex: 2020-06-09T11:51:14).

2. The PSP system will build the digital signature by encrypting the token using the hashing algorithm and the PSP digital certificate private key.

3. The PSP system will convert the digital signature to Base64 String.

4. Make sure that the message formatted with \n only (Unix format), and not \r\n (Windows format).

5. The PSP system will construct the message request XML, which contains the message contents along with the digital signature.

6. The PSP system will submit the message to QMP system using any of the provided channels.

11 Return Reasons

Return Reason Name	Reason Code
Return of goods	50
Invalid paid amount	51
Invalid mobile number	52
Card Transaction Refund	53

12 Mobile Purposes

Purpose Name	Code
Person to Person	1
Person to Public Utility	2
Person to Government Entity	3
Person to Merchant/Business	4
Cash-In	5
Cash-Out	6
Merchant/Business to Person	7
Public Utility to Person	8
Government Entity to Person	9
Merchant/Business to Merchant/Business	10
Remittance	11
Card Transactions	12
Fees	13
Salary	14
Merchant Payment via QPay	15

13 Customer Types

Customer Type	Code
Person	PER
Business	BUSI
Government Entity	GOV
Public utilities	PU
Merchant	MER
Minor	MIN
GCC Visitor	VISGCC
Non GCC Visitor	VISNONGCC
Teller	TELLER
Money Transfer Operator	MTO
Fees Wallet	FEEW
Card Wallet	CARDW
Employer	EMP
Agent	AGT

14 Identification types

Customer ID Type	Code
National ID Number	NIDN
Passport Number	PPT
GCCID	GCCI
Commercial Registration Number	CRNO

15 PSP Information

Sr. #	Code	PSP Name	2-Digit Code	Short Name	Prefix	Alias Suffix
1	QNBAQAQA	Qatar National Bank	01	QNB	QNBA	QNBA
2	CBQAQAQA	The Commercial Bank of Qatar Ltd.	02	CBQ	CBQA	CBQ
3	DOHBQAQA	Doha Bank Ltd.	03	DBL	DBLA	DBL
4	QISBQAQA	Qatar Islamic Bank	04	QIB	QIBA	QIB
5	ABQQQAQA	Al-Ahli Bank	05	ABQ	ABQA	ABQ
6	QIIBQAQA	Qatar International Islamic Bank	06	QIIB	QIIB	QIIB
7	ARABQAQA	Arab Bank Plc.	07	ARAB	ARAB	ARAB
8	MAFRQAQA	Masraf Al-Rayan	17	RYAN	RYAN	RYAN
9	BRWAQAQA	Dukhan Bank	20	DUKH	DUKH	DUKH
10	VODAQAQA	Vodafone	50	VODA	VODA	VODA
11	OREDQAQA	Ooredoo	51	ORED	ORED	ORED